BEYOND INDIVIDUAL CLAIMS: PROBABILISTIC LOGICAL COHERENCE FOR LONG-FORM LLM EVALUATION

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ABSTRACT

Long-form factuality assessment decomposes a response into atomic claims and verifies each against retrieved evidence. The paradigm is effective and now standard, but it embeds an assumption — that the truth of each claim can be settled in isolation — which discards the structure that makes prose an argument. A response can therefore pass per-claim verification while contradicting itself, and two responses built from *identical* claims can differ in coherence because of the order in which they assert them. We formalize *logical coherence* as a property of the joint distribution over claim truth values induced by a Markov random field whose pairwise factors are the relations the response itself draws. The relation vocabulary is a two-level taxonomy: twelve interpretable discourse senses, grounded in PDTB 3.0 and RST, that compile through a deterministic map onto five inferential couplings, each of which fixes a factor table by naming the joint assignment it penalizes. We characterize the factor tables coherence requires by a row-stochasticity property and show that the tables used for claim–evidence factuality are wrong here, collapsing the measure's dynamic range. We then define four candidate coherence readouts over the model — the mean posterior marginal, a two-term consistency probability, a reified coherence node, and a normalized log-partition — give the exact auxiliary-variable construction each requires, and evaluate all four against five stated criteria by exact inference. Only two are monotone in conflict resolution, and the two that fail do so for one shared and instructive reason: they measure whether a conflict is *active* rather than what the reader should *believe*. A third natural candidate is worse still, since a score asking whether asserted relations are honoured is maximized by asserting nothing. Claim priors come from an existing factuality pipeline, making coherence and factuality two stages over one decomposition rather than competing scores. Empirically, on a corpus of response families whose relative coherence ordering is fixed by construction, the readouts recover 82.7% of the declared orderings from gold relation graphs and 34.7–37.1% from graphs mined out of the response text; since the two conditions differ only in the relation graph, the gap localizes the open problem in relation extraction rather than in the probabilistic model.

1 INTRODUCTION

The dominant paradigm for assessing the factuality of long-form generation is decompose-then-verify: split a response into atomic claims, check each against retrieved evidence, and aggregate the verdicts (Min et al., 2023; Wei et al., 2024; Song et al., 2024). It is effective, it is now standard, and it embeds an assumption. Treating each claim as independently checkable discards precisely the structure that distinguishes an *argument* from a *list*.

Consider two failure modes that per-claim verification cannot see. A report may assert both that no passengers were harmed in an incident and that a wire service reported three deaths. One of these claims is true and one is false, and per-claim verification will say so — but it will not register the more basic defect, that the response contradicts *itself*, asserting two things that cannot both hold. Separately, the inferential links a response draws may fail even when their endpoints are individually true: a chain of accurate statements joined by a spurious "therefore" is not a sound argument, and every atom in it verifies. Xie et al. (2026) quantify how often this matters: on logic-related samples, SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of cases respectively.

Factuality and coherence are orthogonal axes. A response can be a bag of true facts that is incoherent, and it can be internally impeccable while being entirely false. The sharpest demonstration in our fixtures is a pair of summaries of one appellate judgment, built from the *same* set of claims — one summary's atom list is a permutation of the other's — and differing only in whether self-serving assertions are stated before or after the findings that qualify them. Every atom is identical, so every per-atom check returns an identical verdict; yet one summary is a faithful account and the other is not, and our measure separates them (0.95 against 0.90). No amount of grounding individual claims against evidence can make that distinction, because the distinction is not about the claims.

Coherence is a statement about a joint distribution. This paper measures the second axis, and the central modelling commitment is that the right object is a joint distribution rather than a tally. Given what a response asserts and how it links its assertions, we ask: how much of what it asserts does the argument, taken as a whole, uphold? That question is answered by a *marginal* in a probabilistic graphical model whose variables are claim truth values and whose factors are the asserted relations — not by counting how many rules a configuration violates. The distinction is not stylistic. A conflict between two claims should propagate: it should depress belief in both endpoints, and through them in whatever those endpoints support. Counting violated rules cannot propagate, and as we show in §5, scoring rule satisfaction directly produces a measure that a response can maximize by asserting no relations at all.

Our contributions are as follows.

- **A coherence MRF and a formal two-level relation taxonomy** (§4). Twelve discourse senses compile through a deterministic map onto five inferential couplings, each defined by the joint assignment it penalizes, which fixes its factor table. Separating the interpretable layer from the inferential one lets the annotation vocabulary be rich while the probabilistic model stays a closed three-parameter pairwise family that no relation type can extend.
- **A characterization of why factuality's factor tables are wrong for coherence** (§4.3). The tables that are correct for claim–evidence factuality charge a premise for being a premise. We characterize the tables coherence requires by a row-stochasticity property, and show the alternative collapses the measure: the root of an entailment chain falls to $\mathbb{P}(a_1{=}1) = 0.19$ and the mean marginal moves only from 0.490 to 0.498 when every conflict edge is deleted, against 0.607 to 0.620 for the correct tables.
- **Four readouts, their exact constructions, and a selection argument** (§5). We give the deterministic auxiliary-variable constructions that make two of them tractable without a $2^{|\mathcal{R}|}$ table, evaluate all four by exact inference across a coherence-ordered family, and establish why the mean posterior marginal is the one to report: it alone satisfies all five criteria, and the two failures share a diagnosable cause (Prop. 3).
- **Two-stage composition with factuality** (§6), taking posterior marginals from an existing factuality pipeline as the coherence model's priors, so one decomposition and one inference stack serve both axes and either can be read alone.
- **Measurements** (§8) that separate relation-extraction error from scoring error by holding the readouts fixed and swapping only the relation graph, plus a worked incoherent/coherent pair with exact values (§9) and a Markov-logic correspondence establishing what the pairwise encoding does and does not buy (Appendix C).

2 Related work

Atomic factuality. FActScore (Min et al., 2023) introduced per-atom precision against a knowledge source, decomposing a biography into atomic facts and reporting the fraction supported. SAFE (Wei et al., 2024) added search-augmented verification with a multi-step reasoning agent per claim and the $F_1@K$ metric that trades precision against a recall target. VeriScore (Song et al., 2024) restricted attention to *verifiable* claims, on the argument that many decomposed atoms are not checkable in principle. FactVerify (Bayat et al., 2023) reasons over retrieved evidence with an undecided verdict class. All four aggregate independent per-claim verdicts, and all four are therefore blind to inter-claim structure by construction.

FactReasoner (Marinescu et al., 2025) replaced verdict counting with probabilistic inference: it builds a factor graph over claim and evidence variables with pairwise factors derived from natural-language-inference relations, and reports posterior marginals rather than a tally. This is the closest antecedent of our model, and we reuse its decomposition, its relation extractor and its inference stack (§3), and take its marginals as our priors (§6). The question it asks is nonetheless different: its factors run between a claim and the *evidence* retrieved for it, so its graph has no claim–claim edges and its marginals cannot express internal consistency. Coherence is a different question asked of the same decomposition.

The logic gap. Xie et al. (2026) states the problem we address: decompose-then-verify pipelines verify atoms individually and overlook the logical dependencies among them, so a text combining true facts through a false dependency is misjudged as factual. Their remedy classifies dependencies as a labelling task over claim pairs. Our departure is to treat the dependencies as *factors in a joint distribution*, which is what allows a local conflict to propagate to distant claims and what makes the resulting score a functional of the whole graph rather than a per-edge average.

Discourse coherence. Entity-grid models (Barzilay & Lapata, 2008) score coherence from the distribution of entity mentions across sentences, and their neural successors (Li & Jurafsky, 2017) learn the same signal with distributed representations. Both are trained and evaluated against *synthetic perturbations* — sentence shuffling and, in later work, contradiction injection — a methodology we extend in §7 by making the perturbation an operator on a declared relation graph rather than on raw text, so its intended effect on the score is known per item. DiscoScore (Zhao et al., 2023) scores generated text using BERT-based discourse representations and correlates well with human coherence judgements. What these approaches share is that coherence is read off surface statistics or learned representations; none exposes an inspectable relation graph, and none yields per-claim attributions. Our relation inventory takes its senses from PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988), but treats them as an interpretable layer that compiles to a smaller inferential vocabulary — which is what makes discourse senses usable as factor types at all.

Consistency checking and LLM judges. SelfCheckGPT (Manakul et al., 2023) detects hallucination by sampling several responses and running pairwise NLI between them; its NLI variant is the natural ablation isolating what global inference adds over purely local contradiction detection, and we propose it as such in §7.1. G-Eval (Liu et al., 2023) is the practical incumbent: an LLM judge scoring a coherence dimension directly from a rubric. Judges of this kind are strong on aggregate correlation and weak on exactly what a graphical model provides — attribution to specific claims, and a guarantee that a meaning-preserving edit cannot change the score.

Relational probabilistic models. Markov logic (Richardson & Domingos, 2006) is the natural alternative encoding, replacing factor tables with weighted first-order rules. Appendix C works out the correspondence and finds it instructive in both directions: the obvious one-clause-per-relation encoding is *not* equivalent to our model and is degenerate at its mode, while a three-clause expansion reproduces it exactly on the three-coupling fragment. We retain the pairwise MRF for the measure because its factor family is closed, and retain the Markov logic view because the beyond-pairwise rules that would address our main modelling limitation are expressible there.

3 STAGE ONE: PROBABILITIES FOR INDIVIDUAL CLAIMS

The coherence model consumes claim-level probabilities and reuses the inference machinery that produces them, so we describe that stage before defining the model itself. It is FactReasoner (Marinescu et al., 2025); the account here is a summary sufficient to make the composition of §6 precise, and it also fixes notation and states the estimator whose label-span construction the relation miner reuses in §4.5.

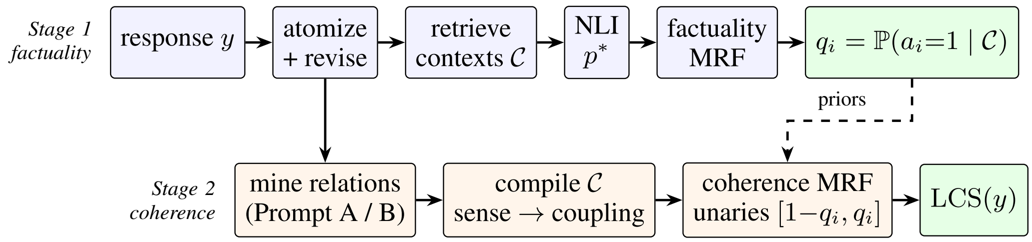


Figure 1: The two-stage pipeline. Stage 1 (§3) is FactReasoner (Marinescu et al., 2025): it grounds each atom against retrieved evidence and returns a posterior marginal q_i. Stage 2 (§4) builds a second MRF over the atoms *alone*, takes q_i as its unary priors, and adds one pairwise factor per mined atom–atom relation. Atomization is shared, so the response is decomposed once. Setting $q_i \equiv \frac{1}{2}$ recovers a pure coherence score that consults no external evidence.

3.1 Decomposition, retrieval, and relation extraction

A response y is decomposed by an LLM *atomizer* into atomic claims $\mathcal{A} = \{a_1, \ldots, a_n\}$, each a single independently verifiable proposition. A *reviser* then decontextualizes each one, resolving pronouns and anaphora so that the claim is checkable outside its original sentence; near-duplicates are collapsed. For each claim a retriever returns supporting passages, which are deduplicated into an evidence set $\mathcal{C} = \{c_1, \ldots, c_m\}$.

An LLM-based NLI extractor then labels admitted pairs. For a claim–evidence pair the label is drawn from $\{\text{entailment}, \text{contradiction}, \text{neutral}\}$; for an evidence–evidence pair both directions are scored and reconciled, with entailment in both directions relabelled *equivalence*. Neutral verdicts are dropped rather than encoded: a claim about which the evidence is silent contributes no factor and is left at its prior, which is what makes the factor set of §4 contain only the relations someone actually asserted.

3.2 Reading a probability off the generation

Each retained relation carries a probability p^*. Rather than asking the model to verbalize a confidence, we read it from the distribution over the tokens it actually emitted. The extractor is prompted to end with a bracketed label; let T be the set of generated tokens overlapping the span of that label in the decoded text. Then

$$p^* \;=\; \exp\Big(\tfrac{1}{|T|} \textstyle\sum_{t \in T} \log p_t\Big), \tag{1}$$

the per-token geometric-mean probability of the label the model produced. Three details of this construction are load-bearing, and we record them because each was a source of silent error.

Span alignment. The probability is measured over exactly the span the label *parser* reports, reconstructing the decoded string from the token list and tracking each token's character offsets. Measuring the same span the label came from makes the label and its probability structurally unable to disagree. Requiring standalone bracket-delimiter tokens instead does not work: tokenizers routinely fuse `[`, `]` and `"` into subwords with adjacent characters, so a delimiter-based rule silently fails on some backends and not others.

Token-array conventions. An earlier version of this code stripped the final entry of the logprob array as an end-of-sequence token. OpenAI-compatible and vLLM logprob arrays contain only emitted content tokens — the stop is signalled out of band — so dropping the last entry deleted a real content token, and for a bracketed label that token is the one closing the label whose confidence is being measured.

Degenerate cases. When the logprob array is empty, no label span is found, or no token aligns to it, the estimator returns 0.5 rather than 0.0. Returning zero would be read downstream as a *degenerate, impossible* relation for a label the model in fact generated, which is a stronger claim than "we could not measure the confidence".

Backends without logprobs. Some serving stacks expose no token logprobs at all. There p^* instead comes from SIMBA-UQ (Bhattacharjya et al., 2025), which samples the extractor repeatedly across a temperature schedule, builds a pairwise similarity matrix over the samples, and aggregates each sample's similarity row into a self-consistency confidence. The winning sample supplies the label and its aggregated confidence supplies p^*, so the downstream model is unchanged.

3.3 The factuality network and its variants

The labelled graph becomes a pairwise MRF with one binary variable per claim and per evidence passage. Unary factors are $\phi_X(x) = [1 - \pi_X,\ \pi_X]$ with π_X the prior on X; each retained relation contributes one pairwise factor from Table 2 below, oriented so the evidence passage is the premise. Marginals $\mathbb{P}(A_i{=}1)$ are computed by weighted mini-bucket elimination (Liu & Ihler, 2011) with i-bound 6, and a claim is reported supported when $\mathbb{P}(A_i{=}1) > \frac{1}{2}$.

Three graph variants differ in which factors are admitted: FR1 connects each claim only to the passages retrieved for it; FR2 deduplicates passages and scores every claim against all of them; FR3 adds evidence–evidence factors. We use FR2. Because scoring every claim against every passage is $O(nm)$ LLM calls, a candidate-pair policy can restrict the pairs actually scored; the calibration of that policy is instructive for the analogous choice in §4.5. Replaying policies against recorded verdicts on a 20-claim narrative with 60 passages, a provenance-restricted policy with an embedding similarity gate holds recall at 1.000 through a threshold of 0.20, slipping to 0.985 at 0.30 and 0.833 at 0.40. Substituting token-Jaccard similarity for sentence embeddings, however, loses 22 of 72 non-neutral relations and moves the factuality score by 0.05 — and *no* threshold recovers full recall, even 0.05. Cheap surrogates for semantic similarity are not safe here, and the saving is workload-dependent rather than constant: roughly $5\times$ on unrelated subtopics but $1.2\times$ on a narrative where every claim shares characters.

Conventions. We use the with-priors factor tables (Table 2) throughout, with $\pi_C = 0.9$ for evidence and $\pi_A = 0.5$ for claims. The published factuality model uses the no-priors tables and a higher evidence prior. The difference is immaterial to the coherence results below, which set claim priors explicitly, but the convention must be stated because the two table families are not interchangeable — that is the subject of §4.3.

4 Stage two: the coherence Markov random field

4.1 The model

Definition 1 (Coherence MRF). *Let $\mathcal{A} = \langle a_1, \ldots, a_n \rangle$ be the atomic claims of a response in assertion order, each a binary variable with $a_i = 1$ read as "the claim holds". Let $\mathcal{R} = \{(s_r, t_r, \tau_r, p_r)\}_{r=1}^{|\mathcal{R}|}$ be a set of* relations*, where $s_r \neq t_r$ index claims, τ_r is an inferential coupling (Def. 2), and $p_r \in [0, 1]$ is the relation's weight. Given priors $\pi \in [0, 1]^n$, the coherence MRF induced by $(\mathcal{A}, \mathcal{R}, \pi)$ is*

$$\mathbb{P}(\mathbf{a}) \;=\; \frac{1}{Z} \prod_{i=1}^{n} \phi_i(a_i) \prod_{r \in \mathcal{R}} \psi_{\tau_r, p_r}(a_{s_r}, a_{t_r}), \qquad Z \;=\; \sum_{\mathbf{a} \in \{0,1\}^n} \prod_i \phi_i(a_i) \prod_r \psi_r, \tag{2}$$

where $\phi_i(a_i) = [1 - \pi_i,\ \pi_i]$ over the states $(0, 1)$ and $\psi_{\tau,p}$ is given by Table 2.

Three structural properties of Definition 1 deserve statement, because each is a deliberate commitment rather than an implementation detail.

Remark 1 (The factor family is closed)**.** *A coupling is nothing more than a choice of which cells of a 2×2 table are down-weighted and by how much. Every $\psi_{\tau,p}$ in Table 2 is determined by p, the source prior π_s, and the penalized set; the whole inferential vocabulary is therefore a three-parameter pairwise family. This is a guarantee about what the model* cannot *do: no relation type, however elaborately named at the discourse level, can smuggle in expressiveness beyond a pairwise potential. Appendix C contrasts this with Markov logic, whose rule vocabulary is open by design.*

Remark 2 (Order-sensitivity enters only through the relations)**.** *Nothing in Eq. 2 references claim position. Assertion order enters exclusively through* which *pairs are related and in* which *direction. It follows that an edit which preserves the relation set — reordering claims whose relations are symmetric, or exchanging a temporal* Precedence *for a* Succession*, which Table 3 maps to no factor at all — provably cannot change any readout. This is a falsifiable prediction about the measure and not merely a design note; §7 describes the ladder built to test it.*

Remark 3 (Coherence is evaluated without evidence)**.** *The graph of Definition 1 contains claim variables only. No retrieved passage appears, and with uniform priors $\pi_i \equiv \frac{1}{2}$ the model consults no external knowledge whatever: it scores how the response hangs together, not whether it is true. This is what makes the two axes separable, and §6 shows how to couple them when a joint reading is wanted.*

4.2 Level one: inferential couplings

Definition 2 (Inferential couplings)**.** *A coupling τ is specified by the set $V(\tau) \subseteq \{0,1\}^2$ of joint assignments (a_s, a_t) it* penalizes *— equivalently, the configurations at which the relation is violated. The vocabulary is:*

Coupling	*Reading*	*Directed*	*Penalized set $V(\tau)$*
ENTAILMENT	$a_s \Rightarrow a_t$	*yes*	$\{(1,0)\}$
CONTRADICTION	$a_s \Rightarrow \neg a_t$	*yes*	$\{(1,1)\}$
EQUIVALENCE	$a_s \Leftrightarrow a_t$	*no*	$\{(0,1),(1,0)\}$
EXCLUSIVE	*exactly one of a_s, a_t holds*	*no*	$\{(0,0),(1,1)\}$
CO_NECESSITY	*at least one of a_s, a_t holds*	*no*	$\{(0,0)\}$
NONE	*no coupling*	—	$\varnothing$ *(no factor)*

Write $\mathcal{T}$ for the five edge-producing couplings and $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$ for the conflict *couplings, those whose penalized set contains the both-true world $(1,1)$.*

Table 1 gives a response excerpt for each coupling, set against the world that coupling denies; since a coupling *is* its penalized set, reading the two together is the quickest way to see what each one commits the response to. The first three couplings are exactly the NLI edge types of §3, which is what lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose, and the distinction between them and CONTRADICTION is the one the taxonomy exists to draw.

Why exhaustiveness needs its own coupling. CONTRADICTION forbids only the both-true world: two claims that cannot both hold, but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what *exhaustive* alternatives assert. "No one was harmed" and "three people died" can neither both hold nor both fail — either someone was harmed or no one was — so a model that treats them as mere opposition has mis-described the situation in a way that matters for scoring: it leaves available a world in which *neither* claim is true, and so under-penalizes. Conversely CO_NECESSITY is the dual, penalizing only the both-false world, and is what a disjunction asserts ("at least one of these two findings must hold"). Adjudicated disputes — incident investigations, litigation, competing hypotheses — are precisely the domains where exhaustive pairs occur, because the function of a finding is to close one. The CONTRADICTION, EXCLUSIVE and CO_NECESSITY rows of Table 1 state one two-part weld failure three times over, so that the only thing varying across them is which same-value worlds the response leaves standing.

Example 1 (A CO_NECESSITY factor in isolation)**.** *Take a single* CO_NECESSITY *relation with $p = 0.9$ over two claims with priors $\pi = 0.5$, and no other factors. From Table 2 the pairwise table is $[1-p,\ \pi_s,\ \pi_s,\ p] = [0.1, 0.5, 0.5, 0.9]$, and multiplying in the unaries and normalizing gives world masses*

$$\mathbb{P}(0,0) = 0.05, \quad \mathbb{P}(0,1) = 0.25, \quad \mathbb{P}(1,0) = 0.25, \quad \mathbb{P}(1,1) = 0.45.$$

The both-false world is suppressed from the 0.25 it would receive under independent fair priors to 0.05, while the three worlds satisfying "at least one holds" share the recovered mass in proportion to how many endpoints they assert. Each marginal rises from 0.5 to 0.7: asserting that at least one of two claims holds is evidence for both, which is the intended reading.

4.3 The pairwise potentials, and why factuality's tables are wrong here

Proposition 1 (Row-stochasticity)**.** *Each with-priors table in Table 2 is row-stochastic in the source state: for $\tau \in \{\text{ENTAILMENT}, \text{CONTRADICTION}, \text{CO\_NECESSITY}\}$ both rows sum to one, giving the*

Table 1: One realistic response excerpt per Level-1 coupling of Def. 2. As in Table 4, each excerpt is prose containing both claims — <u>A is the source, underlined</u>, B **the target, bold**, the *italic* words drawing the link — and $V(\tau)$ is repeated from Def. 2 so the excerpt can be read against the world it rules out. The *Why* column names the world the response denies, which is what distinguishes each coupling from the one above it; the three conflict-adjacent couplings CONTRADICTION, EXCLUSIVE and CO_NECESSITY are illustrated on one two-part weld failure so that only exhaustiveness varies. NONE produces no factor, so its excerpt is one where the text relates the claims without constraining their truth.

Coupling	Reading	$V(\tau)$	Response excerpt, and the world it denies
ENTAILMENT	$a_s \Rightarrow a_t$	$\{(1,0)\}$	"<u>The alloy is chemically identical to the certified reference</u>, *and therefore* **it meets the airframe specification**." *Why:* the response rules out the world where the source holds and the target fails — $(1,0)$ — and says nothing about the case where the alloy differs, which is why the a_s=0 row returns the target to its prior.
CONTRADICTION	$a_s \Rightarrow \neg a_t$	$\{(1,1)\}$	"<u>The weld passed the tensile test</u>, *yet* **the same joint was porous along its full length**." *Why:* only the both-true world $(1,1)$ is denied. Both can still fail together — if the wrong joint was tested, neither claim holds — so $(0,0)$ must stay available.
EXCLUSIVE	exactly one holds	$\{(0,0),(1,1)\}$	"The joint was destructively examined, so *exactly one of* <u>the weld met the tensile minimum</u> *and* **the weld fell short of it** *must be the finding — there is no third outcome*." *Why:* both same-value worlds are denied. The added $(0,0)$ is the whole difference from CONTRADICTION: once the joint has actually been examined the pair cannot both fail, so leaving that world open — as CONTRADICTION does — would under-penalize the conflict.
CO_NECESSITY	at least one holds	$\{(0,0)\}$	"The fracture began at the joint, so *at least one of* <u>the weld</u> *and* **the fastener** *must have given way* — conceivably both." *Why:* the dual of EXCLUSIVE. Only the both-false world is denied, so asserting the disjunction is evidence *for* both endpoints (Example 1).
EQUIVALENCE	$a_s \Leftrightarrow a_t$	$\{(0,1),(1,0)\}$	"<u>The shipment was certified to specification</u> — *that is to say*, **every contractual requirement had been met**." *Why:* the two disagreeing worlds are denied, in both directions; unlike ENTAILMENT there is no vacuous row, since a false source now also forces a false target.
NONE	no coupling	$\varnothing$	"<u>The replacement units were installed in March</u>, *some three months before* **the first in-flight failures were reported**." *Why:* no world is denied. The response genuinely relates the claims, but ordering them in time constrains neither one's truth, so the pair contributes **no factor** and both claims stay at their priors.

Table 2: Pairwise potentials $\psi_{\tau,p}(a_s, a_t)$, listed row-major over the assignments $(0,0),(0,1),(1,0),(1,1)$; π_s is the source claim's prior and bold marks the penalized set $V(\tau)$ of Def. 2. The *no-priors* family is correct for claim–evidence factuality, where the source passage's own truth is at stake; the *with-priors* family is what coherence requires (Prop. 1) and what Def. 1 uses. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

Coupling τ	No priors	With priors	Symmetry
ENTAILMENT	$[p,\ p,\ \mathbf{1-p},\ p]$	$[1-\pi_s,\ \pi_s,\ \mathbf{1-p},\ p]$	asymmetric
CONTRADICTION	$[p,\ p,\ p,\ \mathbf{1-p}]$	$[1-\pi_s,\ \pi_s,\ p,\ \mathbf{1-p}]$	asymmetric
EQUIVALENCE	$[p,\ \mathbf{1-p},\ \mathbf{1-p},\ p]$		symmetric
EXCLUSIVE	$[\mathbf{1-p},\ p,\ p,\ \mathbf{1-p}]$		symmetric
CO_NECESSITY	$[\mathbf{1-p},\ p,\ p,\ p]$	$[\mathbf{1-p},\ \pi_s,\ \pi_s,\ p]$	symmetric

conditional readings

$$\mathbb{P}(a_t{=}1 \mid a_s{=}0) = \pi_s, \qquad \mathbb{P}(a_t{=}1 \mid a_s{=}1) = \begin{cases} p & \tau = \text{ENTAILMENT} \\ 1-p & \tau = \text{CONTRADICTION} \\ p & \tau = \text{CO\_NECESSITY}. \end{cases}$$

Consequently the factor asserts something about the target only when the source is true; when the source is false, the target is returned to its prior.

Proposition 1 is the substantive modelling choice of this section, and the contrast with the no-priors family is worth being precise about. In the no-priors ENTAILMENT table the $(1,0)$ cell is $1-p$ *and the* $(0,\cdot)$ *cells are both* p, so the table charges the *source* being true whenever the target is uncertain. That is appropriate for claim–evidence factuality: there the source is a retrieved passage whose own reliability is in question, and an entailment from a passage to a claim the model doubts is evidence against the passage. It is wrong for coherence, where the source is another claim of the same response, and where a well-supported premise must not be penalized for being a premise.

The consequence is not a matter of degree. On the response of §9.2 — sixteen claims joined by a twelve-edge entailment backbone — the no-priors tables drive the root of the backbone to $\mathbb{P}(a_1{=}1) = 0.19$, far below its 0.5 prior, purely because it is the premise of a chain whose later members are uncertain. The mean marginal sits at 0.490 and moves only to 0.498 when every conflict edge is deleted, a range of 0.008. Under the with-priors tables the same comparison runs 0.607 to 0.620: a range more than 1.5 times larger, from a base that is not already depressed by the backbone itself. All four values are exact, by 2^{16} enumeration. A coherence measure built on the factuality tables does not merely give different numbers; it spends its dynamic range penalizing premises and has little left for conflicts, which are the thing being measured.

4.4 Level two: discourse senses and the compile map

Couplings are the right vocabulary for a factor table and the wrong one for annotation. Asked whether two claims stand in ENTAILMENT, an annotator — human or model — has been handed a question about logic, when the fact actually observable in the text is a question about discourse: does this sentence give a cause, offer evidence, concede a tension, or lay out alternatives? We therefore mine an interpretable layer and compile it down.

Definition 3 (Compile map). *Let $\mathcal{S}$ be the set of Level-2 discourse senses. The compile map*

$$\mathcal{C}:\ \mathcal{S} \longrightarrow \big(\mathcal{T} \cup \{\text{NONE}\}\big) \times \big([0,1] \cup \{\perp\}\big) \times \{0,1\}^3$$

assigns to each sense a coupling, a strength prior ($\perp$ meaning "use the estimator's own value"), and the flags directed, concession *and* ordering-only. *It is total, deterministic, and given in Table 3.*

Table 4 grounds each sense in a response excerpt that contains both claims and the wording that licenses the link, since what an annotator must actually recognize is a form of words rather than a truth-functional pattern. Several entries encode judgements worth defending. Restatement carries a fixed strength prior of 0.90 because a marked paraphrase ("in other words", "equivalently") is

Table 3: The compile map $\mathcal{C}$ of Def. 3. *Ordering-only* senses record assertion order but induce **no factor**, which is what makes an order-only edit provably score-invariant (Rem. 2). Concession is a conflict the text may itself resolve, handled by Eq. 3. The nine senses marked $\star$ form the admissible vocabulary of §4.5.

Level-2 sense	$\mathcal{C}$ coupling	prior	dir.	flag	gloss
Cause-Effect*	ENTAILMENT	$\perp$	yes		source brings target about
Effect-Cause*	ENTAILMENT	$\perp$	yes		source is the effect, target the cause
Evidence*	ENTAILMENT	$\perp$	yes		source is evidence for target
Condition	ENTAILMENT	$\perp$	yes		source conditions target
Instantiation	ENTAILMENT	$\perp$	yes		target instantiates source
Restatement*	EQUIVALENCE	0.90	no		definitional identity
Contrast*	CONTRADICTION	$\perp$	no		opposed, *not* exhaustive
Concession*	CONTRADICTION	$\perp$	yes	conc.	a tension the text resolves
Alternative*	EXCLUSIVE	$\perp$	no		exhaustive: exactly one holds
Disjunction*	CO_NECESSITY	$\perp$	no		at least one holds
Precedence*	NONE	—	yes	ord.	temporal order only
Succession	NONE	—	yes	ord.	temporal order only

a claim about identity of content, and a model asked to estimate its strength has little to add; the estimator's value is used when available, and the prior seeds it otherwise. Five senses collapse onto ENTAILMENT, which is a deliberate loss: the model does not distinguish causation from evidence, because a pairwise potential over truth values cannot. The grouped ENTAILMENT block of Table 4 shows exactly what compilation discards — five distinct discourse facts, one factor table. What survives compilation is the inferential force, and the sense is retained alongside it for interpretability and for error analysis.

The Contrast/Alternative boundary is exactly the exhaustiveness question of §4.2, and the Contrast, Alternative and Disjunction rows of Table 4 are that question in concrete form: one two-part weld failure written up three ways — as mere opposition, as an exhaustive choice, and as a disjunction — differing in nothing but which same-value worlds the response leaves available. The two ordering-only senses are the taxonomy's way of representing a relation the text genuinely draws while asserting that it has no truth-functional content: a narrative that says one event preceded another has related the two claims, and has said nothing about whether either is true.

Concession: a conflict the text resolves. A Concession asserts a tension and then overrides it. "The airline argued pilot error, but the final report found a metallurgical defect" is not an incoherence if the report's finding is presented as authoritative; treating it as a live contradiction would penalize a response for doing the very thing careful argument requires. When a resolving *holding* claim h is present we discount the conflict's weight,

$$p_r^{\text{eff}} \;=\; p_r\,\big(1 - \lambda\,\pi_h\big), \tag{3}$$

with $\lambda = 0.45$; absent a resolver the relation stands at full strength and the tension counts against the response. Holdings are identified by adjudicative cue phrases (*held*, *ruled*, *concluded*, *found that*, *the tribunal*, *ultimately*) on a claim that is not itself an endpoint of the relation and that maximizes lexical overlap with the two endpoints. This is a heuristic, and with λ it is the one place in the design where hand-set constants enter; §10 returns to the point.

4.5 Estimating relations from the response

Definition 1 takes $\mathcal{R}$ as given. In use it must be estimated from the text, and the estimator is the weakest link in the pipeline (§8), so we specify it carefully.

The estimand, decomposed. A relation's weight factors into a type term and a strength term:

$$p_r \;=\; \underbrace{\mathbb{P}(\tau \mid a_s, a_t, y)}_{\text{type confidence}} \cdot \underbrace{\mathbb{P}(a_t \mid a_s, \tau, y)}_{\text{conditional strength}}. \tag{4}$$

The first factor asks how confident we are that this is a τ*-relation at all*; the second, given that it is, how strongly the source determines the target. Conflating them is a mistake: a confidently-identified

Table 4: One realistic response excerpt per Level-2 sense of Table 3, in the same row order and grouped by the coupling each compiles to. Each excerpt is prose of the kind the estimator actually sees and *contains both claims*: A is underlined, B **is bold**, and the *italic* words are the ones that draw the link — the span §4.5 obliges the model to quote, which is why it must be findable in the response and not in either claim. Senses are marked $\star$ when admissible; the three unmarked ones are included because their breadth is the argument for excluding them (§4.5). The Contrast, Alternative and Disjunction rows describe one two-part weld failure three ways, separated only by exhaustiveness (§4.2); they reuse the scenario of Table 1 deliberately, so that a discourse sense and the coupling it compiles to can be compared on the same prose.

Level-2 sense	$\mathcal{C}$	Response excerpt, and why this sense rather than a neighbour
Cause-Effect*	ENTAILMENT	"The company shipped a batch it knew to be out of tolerance, and *as a direct result* **its share price fell** 15% **that quarter**." *Why:* the response itself presents A as bringing B about; the marker names the direction.
Effect-Cause*	ENTAILMENT	"The entire fleet was grounded on 14 March. The order came *because* **inspectors had found a cracked wing spar on two airframes**." *Why:* the same support runs from the stated effect to its cause. Direction of *explanation* is reversed; direction of inferential support is not.
Evidence*	ENTAILMENT	"The maintenance log records no inspection between January and June, *which shows that* **the mandatory 500-hour check was skipped**." *Why:* A is offered in support of B without any claim that it caused it — a record is evidence, not a mechanism.
Condition	ENTAILMENT	"**A part is cleared for airframe use** *only if* its alloy meets the reference specification." *Why:* B is asserted contingent on A. Note how weakly the cue constrains: nearly any dependence can be reread as a condition, which is why the sense is inadmissible.
Instantiation	ENTAILMENT	"Several suppliers failed the 2023 audit — *for instance*, **the Tucson plant failed on two counts**." *Why:* B is one case falling under A's generalization. Like Condition, broad enough to fit almost any pair that shares a topic.
Restatement*	EQUIVALENCE	"The shipment was certified to specification — *in other words*, **every requirement in the contract had been met**." *Why:* a *marked* paraphrase asserts identity of content, which is why the strength prior is fixed at 0.90 rather than estimated.
Contrast*	CONTRADICTION	"The weld passed the tensile test, *but* **the same joint was porous along its full length**." *Why:* opposed and *not* exhaustive. A third possibility — the wrong joint was tested — leaves both false, so the both-false world must stay available.
Concession*	CONTRADICTION	"*Although* the airline argued throughout that pilot error was to blame, **the final report identified a metallurgical defect**; the tribunal *ultimately held* for the latter." *Why:* a tension the text itself settles. The resolving holding triggers the discount of Eq. 3 rather than counting as a live conflict.
Alternative*	EXCLUSIVE	"The two accounts cannot be reconciled: *either* no one aboard was harmed, *or* **three passengers died**, and *exactly one of them is true*." *Why:* exhaustive. Unlike Contrast the pair can neither both hold nor both fail, so the both-false world is penalized too.
Disjunction*	CO_NECESSITY	"Metallurgy confirms the fracture began at the joint, so *at least one of* the weld *and* **the fastener** *must have failed* — possibly both." *Why:* at least one holds and both may. The dual of Alternative: only the both-false world is penalized.
Precedence*	NONE	"The replacement units were installed in March, *some three months before* **the first in-flight failures were reported in June**." *Why:* the text genuinely relates the two claims, and says nothing about either one's truth — so the sense induces **no factor**.
Succession	NONE	"**The first in-flight failures were reported in June**, *fully three months after* the replacement units had been installed." *Why:* the mirror of Precedence — the same two claims, the same absence of truth-functional content. Offering both senses invites an arbitrary choice on a relation that produces no factor either way.

weak causal link and a doubtfully-identified strong one are different epistemic situations that a single elicited number cannot distinguish. Two prompts estimate them separately (full texts in Appendix A).

Prompt A elicits the discourse sense and its coupling, with chain-of-thought *preceding* the label so the model commits to a sense only after reasoning — and so that the label's token logprobs, read exactly as in Eq. 1, measure a decision already made rather than one being made. *Prompt B* elicits the conditional strength as a surrogate token: the answer's first word must be `Yes` or `No`, and the strength is the renormalized $\mathbb{P}(\texttt{Yes})\,/(\mathbb{P}(\texttt{Yes})+\mathbb{P}(\texttt{No}))$ (Kadavath et al., 2022), or the affirmative fraction over N samples where logprobs are unavailable. The question is deliberately graded — "is the implication at least plausible?" rather than "is it certain?" — so that a weak but real link is not forced to answer `No`; its weakness surfaces in the renormalized probability instead. A verbalized variant asking for `[p=0.NN]` directly is retained only as a comparison, verbalized confidence being weakly calibrated (Xiong et al., 2024).

Grounding is mandatory, not optional. Prompt A always receives the full response, and is instructed to assert a coupling only where the response *itself* draws the connection. This is the single most consequential decision in the estimator. Judging a claim pair in isolation invites the model to accept any abstractly plausible relation, and the failure is not a mild over-generation: measured relation density reaches six to nine relations per claim, including spurious contradictions on paragraphs that contain none. Because Eq. 2 is a product over relations, a graph of many soft mutually-constraining edges drives every marginal toward the prior, so an over-connected graph does not merely add noise — it destroys the measure's ability to distinguish a coherent response from an incoherent one, in the direction of calling everything mediocre.

Two further precision mechanisms are used, one requested and one checkable. The *admissibility filter* restricts output to the nine (sense, coupling) combinations marked $\star$ in Table 3. Condition and Instantiation are the notable exclusions: both compile to ENTAILMENT and both are semantically broad — their rows in Table 4 illustrate the breadth, since almost any dependence can be read as a condition and almost any topical pair as an instantiation — so a model reaching for "these two claims are related somehow" lands on them readily; they accounted for 13% of one measured arm's edges while appearing zero times in gold. The *evidence-span requirement* obliges the model to quote, verbatim from the response, the words that draw the link; the parser then verifies that the quotation is actually a substring of the response and drops the relation otherwise. This is the only precision mechanism in the estimator that is *checked* rather than merely asked for. It must quote the response and not the claims, because claims are decontextualized rewrites and only about 27% of them appear in the response verbatim even after normalization.

Two prompt generations. The first-generation Prompt A over-asserts badly: against a corpus in which 7.2% of candidate pairs are genuinely related, it answers "related" on 72% of the pairs a windowed policy hands it. Five properties push it that way, and the revision addresses each. (i) Its few-shot balance was three related to one unrelated; class balance in the exemplars is a strong prior on the answer distribution, and this one pointed the wrong way by an order of magnitude. The revision is two related to three unrelated. (ii) Its single negative exemplar had the response *explicitly disavow* a link ("the two processes were not related this year"), which teaches that NONE requires denial, whereas real prose simply stays silent; the revision's negatives are all silence. (iii) It asked for grounding but gave the model nowhere to show it and nothing to check it — hence the evidence span. (iv) Its first reasoning step was a leading question listing eight affirmative relation types before mentioning NONE; the revision asks the discriminating question first, whether the text draws the link or the claims are merely near each other. (v) It gave NONE one line against nine for relation-bearing senses; the revision gives it first position and equal prose. The revision also names *transitivity* explicitly as a non-relation, since roughly 85% of the first generation's false positives are the model closing a chain the text laid out pairwise, and 96.6% of them touch an endpoint of a real relation. Both generations are retained permanently so that any claim about the prompt can be measured rather than asserted; §8 reports both.

Candidate pairs. Querying all $n(n-1)$ ordered pairs is quadratic in LLM calls and, as argued above, actively harmful. We define four policies:

- ALL_PAIRS: every ordered pair $i \neq j$.

Algorithm 1 SELECTCANDIDATEPAIRS: response-anchored candidate selection.

Require: claims $\mathcal{A} = \langle a_1, \ldots, a_n \rangle$ in assertion order; response y; policy; window w; gate threshold θ_g; discourse thresholds θ_d, δ
Ensure: ordered candidate pairs P, plus a coverage record

```
if policy = ALL_PAIRS then return every ordered pair (a_i, a_j), i ≠ j
W ← {(a_i, a_j) : 0 < j − i ≤ w}                                   ▷ forward order window
if policy = BIDIRECTIONAL then W ← W ∪ {(a_j, a_i) : 0 < j − i ≤ w}
G ← ∅
if policy = GATED then
    G ← {(a_i, a_j) : j − i > w, sim(a_i, a_j) ≥ θ_g}
align each claim to its best-matching sentence of y
Prom ← {(a_i, a_j) ∉ W ∪ G : DISCOURSEADJACENT(i, j)}
Dem ← {(a_i, a_j) ∈ W : ¬ DISCOURSEADJACENT(i, j)}
return (DEDUP(W ∪ G ∪ Prom) \ Dem), (|W|, |G|, |Prom|, |Dem|)
```

- WINDOWED: forward pairs with $0 < j - i \leq w$, default $w = 4$.
- GATED: the window, plus long-range forward pairs whose embedding or entity similarity exceeds a threshold, to catch discourse callbacks.
- BIDIRECTIONAL: *both* arc directions within $|j - i| \leq w$.

The last exists because of a structural limitation, not a tuning preference. A forward-only policy *cannot emit* an arc that runs backward in claim order, so a gold relation that is both directed and backward is unreachable under it no matter what the model says. In our corpus 105 of 603 edge-producing gold relations are of that kind, capping forward-only recall at 82.6% before the model is consulted; §8 measures the effect.

Every policy is then refined against the response itself (Algorithm 1). Each claim is aligned to its originating sentence; a pair is *discourse-adjacent* if the two claims share content tokens above a Jaccard threshold, if their sentences are within a short span, or if the target's sentence opens with a discourse connective. Out-of-window pairs that are discourse-adjacent are *promoted* into the candidate set; in-window pairs that are not are *demoted* out of it. The refinement is a correlated filter — it selects for the same surface adjacency that the sense prompt reads as evidence of relatedness — which is why we report it as a policy component and measure its effect rather than treating it as a neutral preprocessing step.

Failures must be counted, not absorbed. Mining issues one LLM call per candidate pair for Prompt A and one more for each pair that yields an edge, run concurrently under a rate limit. A throttled or failed call returns no parse, and a parser that reads "no parse" as "no relation" converts an infrastructure failure into a sparse graph — which then scores as a *more* coherent response, since fewer conflicts are asserted. We therefore tally call errors separately from genuine NONE answers and report them, and distinguish an explicit NONE from an unparseable coupling for the same reason.

5 FOUR COHERENCE READOUTS

Definition 1 specifies a distribution; a coherence score is a functional of it, and several are defensible. We state five criteria, define four candidates with their exact constructions, and then decide between them by exact inference rather than by preference.

Definition 4 (Criteria)**.** *A coherence readout should be: **(R1)** a genuine functional of the joint* $\mathbb{P}(\mathbf{a})$*, not of the relation list alone; **(R2)** valued in* $[0, 1]$ *and interpretable; **(R3)*** monotone — *resolving or removing a conflict must not decrease it; **(R4)** sensitive to conflict without hand-tuned constants; **(R5)** computable from standard inference queries.*

R1 deserves comment, since it is the criterion that does the most work. A readout that depends only on $\mathcal{R}$ — how many relations there are, what types, what weights — is not measuring the response; it is measuring its own annotation. Proposition 2 below shows this is not a hypothetical concern.

5.1 (A) Mean posterior marginal

$$\mathrm{LCS}_{\mathrm{mm}}(y) \;=\; \frac{1}{n}\sum_{i=1}^{n}\mathbb{P}(a_i=1)\,. \tag{5}$$

The marginals come directly from a single MAR query. The interpretation is the expected fraction of the response's claims that the argument, taken as a whole, upholds. A conflict suppresses the marginals of its endpoints; a satisfied entailment chain propagates support along itself; a claim no relation touches stays at its prior and contributes neutrally.

Alongside the scalar we report three diagnostics that come free with the same query: the per-claim marginals themselves, which localize an incoherence to specific claims; the count of claims dragged *below their own prior*, which is the sharpest single indicator of an active conflict; and the mean normalized binary entropy $\mathcal{E}=\frac{1}{n}\sum_i H_2(q_i)$, lower being more decisive.

5.2 (B) Consistency: a two-term probability

The obvious alternative to averaging marginals is to ask for the probability that the response is consistent. Making that precise turns out to require two terms:

$$\mathrm{LCS}_{\mathrm{cn}}=\tfrac{1}{2}\big(\mathcal{K}+\mathcal{S}\big),\quad \mathcal{K}=\mathbb{P}\Bigg(\bigwedge_{r:\,\tau_r=\text{CONTRADICTION}}\neg\big(a_{s_r}\wedge a_{t_r}\big)\Bigg),\quad \mathcal{S}=\frac{\sum_{r\in\mathcal{R}_{\mathcal{S}}}p_r\,U_r}{\sum_{r\in\mathcal{R}_{\mathcal{S}}}p_r}, \tag{6}$$

where $\mathcal{R}_{\mathcal{S}}$ ranges over $\{\text{ENTAILMENT}, \text{EQUIVALENCE}, \text{EXCLUSIVE}\}$ and U_r is the probability that relation r is *actively upheld*:

$$U_r=\mathbb{P}(a_{s_r}{=}1\wedge a_{t_r}{=}1)\ \text{ for ENTAILMENT, EQUIVALENCE,}\qquad U_r=\mathbb{P}(a_{s_r}\neq a_{t_r})\ \text{ for EXCLUSIVE.}$$

The two coupling sets are deliberately different. The conflict event $\mathcal{K}$ keys on CONTRADICTION alone, even though $\mathcal{T}_{\mathrm{conf}}=\{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$. The reason is that an EXCLUSIVE spreads its incoherence over *both* same-value worlds, so reading only its both-true half would describe half the coupling and silently report the other half as satisfactory. EXCLUSIVE therefore earns credit in $\mathcal{S}$ instead, where its exactly-one world is simultaneously where the exclusion is honoured and where it is informative. CO_NECESSITY enters neither term: its defect is the both-false world, which (a)'s marginals already see directly.

Computing $\mathcal{K}$ exactly. $\mathcal{K}$ is an event probability over a conjunction of $|\{r : \tau_r = \text{CONTRADICTION}\}|$ pairwise events, which a naive encoding would represent as a factor of exponential arity. Instead we augment the network with deterministic auxiliary variables: one u_r per conflict edge with a ternary factor enforcing $u_r = a_{s_r}\wedge a_{t_r}$, then a chained running disjunction $U_k = U_{k-1}\vee u_k$ with ternary factors, and finally $\mathcal{K}=\mathbb{P}\big(U_{|\cdot|}{=}0\big)$. Every auxiliary factor is deterministic and at most ternary, so no $2^{|\mathcal{R}|}$ table is ever built and the augmented network is read with one MAR.

Computing $\mathcal{S}$ exactly. $\mathcal{S}$ is not an event probability but a weighted sum of *pairwise joints* — a linear functional of the joint — so it cannot be read off the same accumulator chain. It needs one deterministic indicator variable per supported edge and its own MAR. Because the indicators are mutually independent given the base network, they may be batched across several queries without approximation, which matters only for inference backends that enumerate.

Why "actively upheld" and not "satisfied". The definition of U_r credits a relation only in the informative world, not in every world where it is technically satisfied. An entailment $a_s \Rightarrow a_t$ is satisfied vacuously whenever $a_s = 0$, and crediting that is fatal:

Proposition 2 (The vacuous-truth trap)**.** *Let $\mathcal{K}'=\mathbb{P}$(every relation in $\mathcal{R}$ is satisfied), the apparently natural extension of $\mathcal{K}$ to all couplings. Then a response with no mined relations satisfies $\mathcal{K}'=1$, the maximum attainable value.*

Proof. $\mathcal{K}'$ is the probability of an intersection of $|\mathcal{R}|$ events; for $\mathcal{R}=\varnothing$ the intersection is the whole space. □

The proposition is immediate, but its consequence is not, and it is the reason $\mathrm{LCS}_{\mathrm{cn}}$ has the form it does. Measured on the family of §9.2, a disconnected bag of claims scores $\mathcal{K}' = 1.000$ while a response whose causal spine is fully satisfied scores 0.221 — an inversion, not a tie. Nor is the defect repaired by count normalization: the geometric mean over relations gives 0.883 for the base response against 0.882 for the coherent one, non-monotone *and* still 1.000 on the bag; the expected fraction of satisfied relations gives 0.857 against 0.852, likewise inverted; and normalizing against a prior ceiling leaves $[0, 1]$ altogether (8.82 against 5.46) while still running backwards. The common cause is that "are the asserted relations honoured?" is a property of $\mathcal{R}$ rather than of y — a violation of R1 — and it is maximized by asserting nothing. Crediting only the informative world removes the free lunch: a bag of claims upholds nothing and scores $\mathcal{S} = 0$.

The two terms are averaged rather than multiplied so that neither can annihilate the other. A live contradiction should not erase a well-supported spine, and a relation-free response should not inherit a perfect score from an empty conflict set.

5.3 (C) A REIFIED COHERENCE NODE

The most direct reading of "is this response coherent?" as a single probability is to introduce a variable for it. Add a binary R with Bernoulli prior ρ and, for each relation, a ternary *vote* factor

$$h_r(R, a_s, a_t) = \begin{cases} 1 & R = 0 \quad \text{(the branch is flat)} \\ 1 & R = 1,\ (a_s, a_t) \notin V(\tau_r) \\ 1 - p_r & R = 1,\ (a_s, a_t) \in V(\tau_r), \end{cases} \qquad \mathrm{LCS}_{\mathrm{re}} = \mathbb{P}(R = 1)\,. \tag{7}$$

In the $R = 1$ branch each violated relation charges $1 - p_r$, so R behaves as a noisy-AND over relation satisfaction; in the $R = 0$ branch nothing is charged, so $R = 0$ carries no commitment. With no relations R is decoupled and $\mathrm{LCS}_{\mathrm{re}} = \rho$, correctly signalling that nothing was measured. Note that the violated set is $V(\tau)$ from Def. 2, so ENTAILMENT is treated as satisfied at $a_s = 0$ — vacuously — which is where (c) inherits a weaker form of Prop. 2's problem. Appendix B works a three-claim subnet through by hand, giving $\mathbb{P}(R{=}1) = 0.453$.

5.4 (D) NORMALIZED LOG-PARTITION

The partition function measures how much probability mass the constraints leave standing, which is a global coherence signal requiring no marginals at all. It is unnormalized, so we grade it between two references built from the same edge skeleton:

$$\mathrm{LCS}_{\mathrm{lp}} \;=\; \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}. \tag{8}$$

For the ceiling, $Z_{\max}$ is the partition function of the same network with every conflict edge removed: deleting constraint factors only adds mass, so $\log Z \leq \log Z_{\max}$ always. For the floor we take the *MAP world mass* of the base network, $Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$.

Lemma 1 (The MAP world mass is a valid, coherence-graded floor)**.** *$\log Z_{\min} \leq \log Z$ for every network, and $Z_{\min}$ is non-increasing in conflict strength.*

Proof. $Z = \sum_{\mathbf{a}} \mathrm{mass}(\mathbf{a})$ is a sum of non-negative terms, so its single largest term is a lower bound. For the second claim, increasing p_r on a conflict edge multiplies the mass of every world in $V(\tau_r)$ by a factor below one and leaves other worlds unchanged, so the maximum over worlds cannot increase. □

Lemma 1 is not a technicality; it replaces two floors that look more natural and are wrong. Retyping every edge to CONTRADICTION, or saturating conflicts to $p = 1$, both fail to bound $\log Z$ below for the row-stochastic tables of Prop. 1: an isolated conflict removes no mass at all under those tables, whereas a mass-concentrating entailment backbone concentrates a great deal, so a network combining a strong backbone with several soft conflicts can have a base $\log Z$ *below* either candidate floor. We observed exactly this on real graphs, which put Eq. 8 outside $[0, 1]$ until the MAP floor replaced them. In practice we clamp the ratio to $[0, 1]$ regardless, since approximate inference computes $\log Z$ and the MAP bound with finite i-bound and small tolerance violations are possible.

Table 5: All four readouts across a coherence-ordered family of five variants of one response, by exact inference over 2^{16} worlds. The variants run from least to most coherent left to right in the intended ordering. Bold marks a value that moves the *wrong* way relative to its predecessor. Only (a) and (d) are monotone throughout.

Variant	(a) $\mathrm{LCS}_{\mathrm{mm}}$	(b) $\mathrm{LCS}_{\mathrm{cn}}$	(c) $\mathrm{LCS}_{\mathrm{re}}$	$\log Z$	(d) $\mathrm{LCS}_{\mathrm{lp}}$
worse	0.586	**0.778**	0.116	−10.54	0.68
base	0.607	0.781	0.148	−9.64	0.78
concession	0.612	**0.772**	**0.134**	−9.64	0.80
fixcasualty	0.613	**0.764**	0.156	−8.95	0.89
coherent	0.620	**0.752**	0.181	−8.25	1.00
monotone?	**yes**	no	no	**yes**	**yes**

Table 6: The four readouts against the criteria of Def. 4, with the inference queries each requires.

Readout	Queries	R1	R2	R3	R4	R5	Note
(a) mean marginal	MAR	✓	✓	✓	✓	✓	**selected**
(b) consistency	MAR×2	✓	✓	×	✓	✓	concession dip
(c) reified	MAR	✓	✓	×	×	✓	free ρ; dip
(d) norm. $\log Z$	PR, MAP	✓	✓	✓	✓	✓	needs 2 references

5.5 Deciding between the four

Table 5 settles the question empirically and Table 6 summarizes. Readouts (b) and (c) both fail R3, and they fail it at the *same* variant — the one where a conceded tension is resolved — for a shared reason that is worth stating precisely, because it looks like an implementation bug and is not.

Proposition 3 (Belief versus event activity)**.** *Softening a resolved conflict via Eq. 3* increases *the probability of its both-true world. For a single conflict edge with uniform priors, discounting p from* 0.80 *to* 0.55 *moves* $\mathbb{P}(a_s{=}1, a_t{=}1)$ *from* 0.10 *to* 0.225*.*

Proof. Under both conflict couplings the $(1,1)$ cell of Table 2 is $1-p$, which increases as p falls. Take a lone CONTRADICTION edge with $\pi_s = \pi_t = \frac{1}{2}$, so the with-priors table is $[\frac{1}{2}, \frac{1}{2}, p, 1-p]$ and each world carries an equal unary factor of $\frac{1}{4}$. The unnormalized masses are then proportional to the table itself, giving $\mathbb{P}(1,1) = (1-p)/\big(\frac{1}{2}+\frac{1}{2}+p+(1-p)\big) = (1-p)/2$, which is monotone decreasing in p. Evaluating at $p = 0.80$ and $p = 0.55$ gives 0.100 and 0.225. □

The consequence sorts the four readouts into two kinds. A readout that asks *is the conflict active* — (b) counts exactly that event, (c) charges $1-p_r$ for it — must register a softened conflict as *more* often active, and therefore dips. A readout that asks *how strongly should each claim be believed* — (a) through marginals, (d) through mass — sees less suppression and rises. Since resolving a concession is by construction an increase in coherence, a coherence score must track belief rather than event activity. That is the substantive finding of this section, and it is not specific to our parameterization: any readout defined as the probability of a conflict-free configuration will invert under any discount-based treatment of resolved concessions.

A second failure mode of (b). Table 5 shows (b) failing worse than a single dip, and the reason is instructive about $\mathcal{K}$ and $\mathcal{S}$ jointly. In this family both conflicts are EXCLUSIVE rather than CONTRADICTION, so $\mathcal{K} \equiv 1.000$ throughout by the coupling-set decision above, and the score is carried entirely by $\mathcal{S}$. But removing an exclusion removes *support* credit as well as conflict, so the score runs backwards from 0.778 to 0.752 across the whole ladder. On three-coupling variants of the same response, where the conflicts are CONTRADICTION and $\mathcal{K}$ is live, (b) instead runs $0.576 \to 0.654 \to \mathbf{0.582} \to 0.672 \to 0.752$: correctly ordered apart from the concession dip. Either way it is non-monotone, and the lesson is that $\mathcal{K}$ and $\mathcal{S}$ should be read separately as diagnostics rather than averaged into a headline.

Algorithm 2 $\textsc{ComputeLCS}(y)$ — the end-to-end coherence score.

Require: response y; relation estimator $\mathcal{M}$; discount λ; priors π
Ensure: LCS $\in [0, 1]$, marginals $\{q_i\}$, diagnostics
1: $\mathcal{A} \leftarrow \textsc{AtomizeAndDecontextualize}(y)$
2: $P \leftarrow \textsc{SelectCandidatePairs}(\mathcal{A}, y, \dots)$ ▷ Algorithm 1
3: $\mathcal{R} \leftarrow \varnothing$
4: **for all** $(a_i, a_j) \in P$ **do** ▷ concurrent, rate-limited
5: $\quad$ $(\text{sense}, \tau, c) \leftarrow \textsc{PromptA}(a_i, a_j, y)$ ▷ $c = \mathbb{P}(\tau \mid \cdot)$ from Eq. 1
6: $\quad$ **if** $\tau = \textsc{none}$ **then continue**
7: $\quad$ $\sigma \leftarrow \textsc{PromptB}(a_i, a_j, \tau, y)$ ▷ surrogate-token strength
8: $\quad$ $\mathcal{R} \leftarrow \mathcal{R} \cup \{(i, j, \tau, \ \mathrm{clip}_{[0,1]}(c \cdot \sigma))\}$
9: **for all** $r \in \mathcal{R}$ with $\tau_r \in \mathcal{T}_{\mathrm{conf}}$ sensed as Concession **do**
10: $\quad$ **if** a resolving holding h exists **then** $p_r \leftarrow p_r(1 - \lambda \pi_h)$
11: $\quad$ **else** mark r unresolved
12: $\mathcal{N} \leftarrow$ network of Eq. 2 over $(\mathcal{A}, \mathcal{R}, \pi)$ ▷ with-priors tables
13: $\{q_i\} \leftarrow \textsc{Infer}(\mathcal{N}, \textsc{mar})$
14: **return** $\frac{1}{n} \sum_i q_i$, $\{q_i\}$, $\big(\#\{q_i < \pi_i\},\ \mathcal{E},\ \log Z\big)$

Selection. We report (a) as the headline. It satisfies all five criteria, needs a single MAR query, and is the only one of the four whose value decomposes into per-claim contributions — which is what makes an incoherence *locatable* rather than merely detectable. We compute and report all four, since (d) is a genuine alternative satisfying the same criteria through a different route (mass rather than marginals), and $\mathcal{K}$, $\mathcal{S}$ carry diagnostic information the mean marginal does not expose.

5.6 Algorithm and cost

Algorithm 2 is dominated by steps 4–9: $|P|$ Prompt-A calls plus one Prompt-B call per surviving edge, which the candidate policy keeps near-linear in n rather than quadratic. Inference is negligible by comparison.

Answering all four readouts requires seven inference calls: the base MAR and PR, the $\mathcal{K}$ accumulator MAR, the $\mathcal{S}$ indicator MAR, the reified MAR, the conflict-free ceiling PR, and the base MAP. Computed independently they would cost fourteen, since each readout re-runs the shared base pair. Production inference is weighted mini-bucket (Liu & Ihler, 2011) with i-bound 6; every *exact* value reported in this paper instead comes from explicit 2^n enumeration, which we use as a regression oracle for $n \leq 20$ and which is the source of Tables 5 and the values in §9.

6 Composing coherence with factuality

The priors π of Definition 1 are a free slot, and the natural thing to put in them is what Stage One already computed (Figure 1):

$$\pi_i \ \longleftarrow \ q_i \ = \ \mathbb{P}(a_i = 1 \mid \mathcal{C})\,. \tag{9}$$

A claim is then credited when it is both externally supported and internally upheld, and the two readings compose in the natural direction: an unsupported claim enters the coherence graph already doubted, so an argument resting on it is penalized twice — once for the evidence and once for what the argument does with it. Setting $q_i \equiv \frac{1}{2}$ recovers the pure coherence score of Remark 3, so the composition is an option rather than a commitment and the two axes remain separately reportable.

Two stages rather than one joint network. One could instead build a single MRF over claim and evidence variables with claim–claim edges added. We do not, for three reasons. Inference runs over n variables rather than $n(1 + k)$ for k retrieved passages per claim, which matters because the augmented networks of §5.2 add further variables on top. The mined relation graph can be rescored under several prior sets — uniform, factuality-derived, ablated — without re-running the estimator, which makes prior ablations essentially free. And the factuality stage stays independently cacheable and independently reportable, which a fused model forfeits. Priors are applied at scoring time and never by mutating the mined graph, so one mined graph supports every prior condition.

Aligning claims across stages. The two stages must agree on which claim is which, and the obvious key is the wrong one. Matching proceeds by object identity first (the shared-atomization path, where the response is decomposed once and both stages see the same objects), then by normalized text (casefolded, whitespace-collapsed, terminal punctuation stripped), and only then by identifier. Text precedes identifier deliberately: near-duplicate removal keeps the first-seen claim and leaves the surviving identifier set *sparse* — $a_0, a_1, a_3, \ldots$ — so two independent decompositions of the same response can disagree about which claim occupies which index. Matching on identifiers alone would then attach a prior to the *wrong* claim, which is strictly worse than attaching none, since it silently corrupts the score rather than degrading it. When coverage falls below one half the run is flagged degraded, and the configured policy is to warn, to raise, or to discard every prior and fall back to a clean coherence-only score rather than report a half-primed mixture.

7 Evaluation protocol

Validating a coherence measure requires responses that differ in coherence while holding content fixed. Natural corpora do not supply this: two documents that differ in coherence almost always differ in topic, length and claim set as well, so any score difference is unattributable. Human coherence ratings supply a target but not a controlled contrast. And no existing resource supplies what separates *scoring* error from *estimation* error — a gold relation graph, against which the estimator of §4.5 can be measured directly.

We therefore evaluate on synthesized *coherence ladders*. A ladder is a family of five responses over one claim set: a base response realizing a declared relation plan, and four variants produced by applying perturbation operators to that plan — adding a spurious relation, resolving a conceded tension, dropping conflicts — each variant carrying *its own* gold relation graph derived from the base graph by the same operators. Each family declares the ordering its rungs must exhibit, in three classes: **C1**, a strict increase between consecutive rungs; **C2**, a *predicted* decrease or invariance — which is where Prop. 3's concession dip becomes a falsifiable prediction rather than an excuse, and where an ordering-only edit must be exactly flat by Remark 2; and **C3**, endpoint separation between the extreme rungs. A measure is scored on the fraction of declared constraints it satisfies.

Arms. An *arm* fixes where the relation graph comes from while holding the claims, the priors and the readouts identical. The GOLD arm takes each item's own labelled relations; GOLD_VALID keeps only those labelled valid; a MINED arm runs the estimator of §4.5 on the response text. This is the central experimental device of the paper. Because the readouts are byte-identical across arms, any shortfall of a mined arm below the gold arm is attributable to relation estimation and not to the probabilistic model, and the two error sources never have to be disentangled after the fact.

Match levels. Mined relations are compared to gold at three nested levels: **pair** (were these two claims related at all), **coupling** (pair, plus the Level-1 coupling, with direction required for the asymmetric couplings), and **sense** (plus the Level-2 sense). The coupling level is the one that matters, since it is the level at which Table 2 selects a factor, and it is materially stricter than pair level.

7.1 Baselines

A coherence measure should be compared against controls chosen to fail in informative ways. We describe five families and state what each is diagnostic of. Four are implemented and share the LCS's own decomposition — each baseline is handed the same atoms the LCS was scored on and, like the coherence MRF, consults no retrieved evidence — so the comparison is an ablation rather than a separate experiment. Metrics requiring a source document or a reference are therefore excluded by construction, and we say so where the exclusion is not obvious.

1. *Ladder-blind controls*: claim count, relation count, response length, and number of perturbation operators applied. These cost nothing and they catch the most likely way a coherence measure is secretly wrong — by tracking edit count or graph density rather than coherence. A measure that cannot beat them on C3, or that fails to stay flat where they do, has not earned its complexity. Steen & Markert (2022) make the same argument for coherence measures generally, and introduce intra-system correlation and bias matrices to identify measures whose apparent performance

comes from system-level confounders; we adopt the controls in that spirit. A control that *succeeds* is bad news for the item, not good news for the control, and §7.2 reports one such case.

2. *Local contradiction detection*: pairwise NLI over claim pairs with no global model, scored as one minus the fraction of pairs labelled contradictory, in the spirit of SelfCheckGPT's NLI variant (Manakul et al., 2023). This isolates precisely what joint inference adds over local detection. The prediction is that it does creditably where the defect is a *present* contradiction and poorly where the defect is a *missing* entailment, since a broken chain has no contradictory pair to find. We run it with the same NLI extractor as §3, over the same atoms, so the only differences from the LCS are relation *typing* and *propagation*.

3. *Structured reasoning consistency*: ROSCOE's Self-Consistency (Golovneva et al., 2023), $1 - \max_i \max_{j<i} p_{\text{contr}}(h_i, h_j)$, which is evidence-free like ours but differs in three separable ways — it is a *maximum*, so a second conflict changes nothing; it is *forward-only*, so a relation running backward in claim order is unreachable (§4.5 measures 105 of 603 gold relations as both directed and backward); and it is *untyped*. We therefore also report a mean-aggregated and a symmetric variant, which turn one baseline into an ablation identifying *which* of the three differences accounts for any gap. ROSCOE targets step-by-step reasoning chains, and a long-form response is not a chain, so we label it adapted.

4. *LLM judges*: G-Eval's coherence dimension (Liu et al., 2023), with the SummEval coherence rubric (Fabbri et al., 2021), and a direct "rate the logical coherence from 1 to 5" prompt as the no-rubric control. These are the practical incumbent and the comparison reviewers will want. The ladder is a demanding test for them, since consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while remaining flat where meaning is preserved. We report each judge as a mean over five runs with its standard deviation, because the determinism of Remark 2 is only an advantage next to a measurement of the judge's instability, and because a spread wider than the gap being discussed bounds what the comparison can conclude.

5. *Discourse representations*: DiscoScore (Zhao et al., 2023), with an entity-grid model (Barzilay & Lapata, 2008) as the classical reference. DiscoScore is presented as reference-based and most of its variants consume a reference, which a standalone response does not have; three of them (RC, LC and ENTITYGRAPH) ignore the reference argument entirely and score a single document, and those are the three we use — the last of which *is* an entity grid, so the classical reference comes from the same implementation. All three reduce to noun-lemma repetition across sentences, with no representation of entailment or contradiction, which makes them a sharp test of whether coherence is separable from *cohesion* rather than a weak competitor: §7.2 reports a pair of responses with identical nouns and identical repetition structure, one self-consistent and one self-contradictory, on which all three are exactly flat.

6. *Internal ablations*: uniform versus factuality priors (Eq. 9); the four readouts of §5; the four candidate-pair policies; the two Prompt-A generations; and the gold-relation ceiling. These separate the model's design choices from the estimator's accuracy.

Beyond ladder ordering we recommend Spearman correlation against human coherence ratings on the same items, reported alongside per-family ordering agreement. Ordering agreement is the sharper instrument, being a controlled contrast with a known answer; human correlation is what establishes that the construct being ordered is the one readers care about. Neither substitutes for the other.

A baseline we cite but do not run. The closest existing work to our relation-mining stage is Liu & Strube (2025), who also build a graph over sentence pairs labelled with PDTB 3.0 senses — via a discourse parser rather than an LLM — and add entity links. We do not report a head-to-head number, and the reason is a difference in output type rather than a convenience: their model emits a *three-way* label (low, medium, high), and a three-way label cannot be checked against the ordering constraints of §7, which ask whether consecutive rungs move in a declared direction. The comparison would also require their trained classifier and a PDTB parser, neither of which is a pretrained scorer one can apply to a new response. Three differences are worth recording regardless, because they are the axes on which the two approaches diverge: their model performs no probabilistic inference, so a local conflict cannot propagate; it retains only forward-order pairs ($i < j$), so a backward relation is unreachable, which is the same structural cap we measure in §4.5; and it produces no per-claim attribution, so an incoherence is detected but not located.

7.2 What the baselines show

Two properties of these baselines are worth establishing before any ladder sweep, because both bear on what the comparison in §8.5 can support.

First, *cohesion is not coherence, and the discourse floor cannot tell the difference.* On a pair of two-sentence responses with identical nouns and identical repetition structure — "The weld passed the tensile test. The weld held under load." against "… The weld *failed* under load." — all three single-document DiscoScore metrics return *exactly* equal values, while the pairwise-NLI and ROSCOE baselines separate the pair. The metrics are defined over noun repetition, so this is a property of their definitions rather than an accuracy failure; it is also the cleanest available demonstration that the axis we measure is not the one entity-based coherence models measure.

Second, *a controlled contrast has to be checked, not assumed.* A measure can reproduce a declared ordering for reasons that have nothing to do with coherence, so a contrast is only informative once the model-free proxies have been shown *not* to reproduce it. The claim-identical ordering pair of §9.3 is the clean case: claim count is exactly flat there by construction, response length differs by four tokens (265 against 269) in the direction *opposite* to the declared ordering, and all three discourse metrics rank the adversarial summary *above* the faithful one, the self-serving-first ordering carrying more noun repetition. A cohesion metric prefers the less coherent response of the two. The ladder corpus of §8 generalizes exactly this property: every rung of a family shares one claim set, so a claim-count proxy is flat by construction across all seventy items.

8 Experiments

8.1 The corpus

Evaluation is on a single body of data: a generated corpus of coherence ladders. §7 defines what a ladder, an arm and a match level are; this subsection gives the concrete instance.

The corpus holds **fourteen families and seventy items** — fourteen ladders of five rungs, each rung a distinct response over one claim set. It was produced by `claude-opus-5` and admitted by a three-model cross-vendor validation committee (`gpt-5.6-terra`, `gemini-3.1-pro`, `claude-sonnet-5`), which recovers the relation graph from prose without seeing the plan (V1, any-of), audits the prose for leakage and hedging (V3, majority) and checks atom coverage (V4, majority). Items carry 15–16 claims and 609–698 words, for 1,115 atoms and 670 gold relations in total at a 0.61 valid fraction, together with 342 claim pairs the corpus *explicitly declares unrelated* — a labelled negative set rather than an unlabelled remainder. Subject matter spans fourteen canonical topics across all four domains, so no single topic or domain carries a result.

All four ladder types are represented, and the mix matters for reading the results: eight CONFLICT families (rungs resolve or remove conflicting relations), two CHAIN (entailment links are broken and repaired), three ORDER and one CONTROL. The last two are the negative controls. Every rung of an ORDER or CONTROL family applies only meaning-preserving edits — sentence reordering, or a PRECEDENCE↔SUCCESSION swap that couples no truth values — so those ladders assert *invariance*, and a measure that trends across them is tracking surface form rather than coherence. CONFLICT and CHAIN assert a monotone increase. A corpus with only the increase-type ladders cannot distinguish a coherence measure from an edit-count proxy, which is why both kinds are present.

8.2 Protocol

Each family declares the ordering its rungs must exhibit, per readout, and those declarations are the unit of measurement: 202 **constraint checks** in total, decomposing as 112 over CONFLICT, 30 over CHAIN, 45 over ORDER and 15 over CONTROL. A check names a readout and a rung pair and asserts increase or invariance; a missing score fails rather than passes.

An arm fixes where the relation graph comes from and holds everything else — the claims, the atom priors, the readouts, the scorer settings — byte-identical, so a difference between two arms is attributable to relation sourcing alone. The GOLD arm compiles each item's own labels; GOLD_VALID drops the deliberately-planted invalid ones. The mined arms run the estimator of §4.5 over the response prose with two miners, `llama-3.3-70b-instruct` and `gpt-oss-120b`, both read

Table 7: Mined-versus-gold agreement over the 603 edge-producing gold relations (670 total, less the ordering-only PRECEDENCE/SUCCESSION edges, which compile to Level-1 `none` and contribute no factor), micro-averaged over seventy items. **Fwd/Bwd** split coupling-level recall by whether the gold relation runs forward or backward in claim order; 105 of the 603 run backward. **Non-rel** counts violations of the 342 pairs the corpus explicitly declares unrelated — a real negative set, unlike the unlabelled remainder. **Dup** counts unordered pairs carrying two edges, which BIDIRECTIONAL emits by construction and which place two factors on one variable pair.

Miner	Policy	Edges	Pair R	Coup P	Coup R	Coup F1	Fwd	Bwd	Non-rel
`llama-3.3-70b`	WINDOWED	1106	82.3	34.1	62.5	44.1	71.1	21.9	34/342
`llama-3.3-70b`	BIDIRECTIONAL	3172	91.7	17.9	79.3	29.3	79.9	76.2	119/342
`gpt-oss-120b`	WINDOWED	1094	83.4	40.2	73.0	**51.9**	82.1	29.5	38/342
`gpt-oss-120b`	BIDIRECTIONAL	2973	**96.5**	22.4	**88.4**	35.8	**89.0**	**85.7**	147/342

through token logprobs, under two candidate-pair policies: WINDOWED (forward-only, within an order window) and BIDIRECTIONAL (both arc directions within the same radius).

Mining is sampled once per cell at the model's default temperature, so the mined arms are not deterministic. We measure that rather than estimate it: two *independent* runs of the same two WINDOWED configurations moved every pair, coupling and sense F1 by at most 0.014, while the gold arms reproduced bit-for-bit across the same pair of runs. Per-arm ladder agreement is the noisier statistic, moving by four to six of 202 assertions between those runs, so a single arm's percentage should not be read to the point. That 0.014 supersedes the ± 0.06 we previously estimated on a ten-item corpus: the larger corpus tightens reproducibility rather than merely enlarging the denominator. These are measurements, not significance claims. All exact values are by 2^{16} enumeration; all corpus-scale values are by weighted mini-bucket with i-bound 6.

8.3 RELATION ESTIMATION IS THE BOTTLENECK

Table 7 measures the estimator of §4.5 against gold, and three findings stand out.

First, the forward-only policy is *structurally* capped, exactly as predicted, and the cap is now measured on a corpus large enough to see it clearly: WINDOWED recovers 21.9–29.5% of backward-running gold relations against 76.2–85.7% for BIDIRECTIONAL, because 105 of the 603 scorable gold relations are both directed and backward and no forward-only candidate set can emit them. This is a property of the candidate policy, not of either model.

Second, admitting both arc directions buys that recall at a precision cost severe enough to invert the F1 ordering. BIDIRECTIONAL roughly triples the emitted edges (1,106 $\rightarrow$ 3,172 for `llama`), lifts coupling recall by 16.8 points, and *halves* coupling precision (34.1 $\rightarrow$ 17.9), so coupling F1 falls from 44.1 to 29.3. The mechanism is the one §4.5 anticipated for ALL_PAIRS and it appears here in milder form: because Eq. 2 is a product over relations, spurious factors drive marginals toward the prior rather than merely diluting a signal. The 1,231 duplicate unordered pairs are the same defect seen structurally — two factors on one variable pair, which the model has no principled way to combine — and declared-non-relation violations rise from 34/342 to 119/342 in step.

Third, the two miners differ in the direction that matters, and `gpt-oss-120b` is the stronger relation estimator at every level of the taxonomy: on WINDOWED it beats `llama-3.3-70b` by 7.8 points of coupling F1 (51.9 vs. 44.1), 9.3 points of sense F1, and 10.5 points of backward recall, while emitting slightly *fewer* edges. The gap widens as the label gets finer, which is what a genuine capability difference looks like rather than a threshold artefact: both models find roughly the same pairs (pair recall 82.3 vs. 83.4) and then disagree about what those pairs mean.

A caveat belongs with that third finding, because it cuts against the natural reading. The `gpt-oss` strength estimates are almost entirely saturated: 99% of its edge strengths sit at the numerical boundary (25% at the floor, 75% at the ceiling) against 58% genuinely interior values for `llama-3.3-70b`. Probed directly, the model answers the strength surrogate with all of its top-five mass on one side for easy pairs and with values as small as 10^{-9} for hard ones — so this is the model's own near-determinism, not a defect of the reader. `gpt-oss` therefore buys its accuracy advantage while supplying almost no graded uncertainty, and `llama` is the reverse. A practitioner who needs

Table 8: Left: declared ordering constraints satisfied, grouped by ladder type — 202 assertions per arm in total (8×14 CONFLICT, 2×15 CHAIN, 3×15 ORDER, 1×15 CONTROL). Right: each readout averaged over the seventy items. ORDER and CONTROL assert *invariance*: every rung of those ladders applies only meaning-preserving edits (sentence reordering, or a PRECEDENCE↔SUCCESSION swap), which move no factor, so a readout that trends there is tracking surface form rather than coherence. GOLD_VALID is near-flat on CONFLICT by construction — a conflict-fixing rung removes the planted-invalid conflicts this arm has already discarded — so its failures carry no information about the readouts; the GOLD arm is what the constraints test. Averaging a readout across the rungs of a ladder deliberately collapses the ordering the ladder exists to test, so the right-hand block is a level comparison only.

		Constraints satisfied					Mean readout value			
Arm	Policy	Cfl	Chn	Ord	Ctl	%	LCS_{mm}	LCS_{cn}	LCS_{re}	LCS_{lp}
GOLD	—	89/112	18/30	45/45	15/15	**82.7**	0.795	0.515	0.238	0.410
GOLD_VALID	—	37/112	12/30	45/45	15/15	54.0	0.794	0.576	0.316	0.485
mined by `llama-3.3-70b`										
	WINDOWED	53/112	17/30	0/45	0/15	34.7	0.796	0.676	0.384	0.224
	BIDIRECTIONAL	57/112	18/30	0/45	0/15	37.1	0.723	0.542	0.205	0.037
mined by `gpt-oss-120b`										
	WINDOWED	59/112	12/30	0/45	0/15	35.1	0.825	0.500	0.425	0.129
	BIDIRECTIONAL	53/112	17/30	0/45	0/15	34.7	0.808	0.462	0.251	0.033

calibrated edge probabilities rather than accurate edge labels should read this table in the opposite direction.

8.4 THE MODEL ORDERS CORRECTLY WHEN THE GRAPH IS CORRECT

On gold relations the readouts satisfy 82.7% of the declared ordering constraints — 167 of 202 (Table 8). Mined arms reach 34.7–37.1%. Since the readouts, the claims and the priors are byte-identical across arms, the 46–48 point gap is attributable to relation estimation alone, and it is consistent with Table 7: at 29–52% coupling F1 roughly half the factors in a mined network are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper's central empirical claim, and it is a claim about *where the remaining problem is*: the probabilistic model orders responses correctly when handed a correct graph, and the open problem is extracting one.

The per-ladder breakdown localizes the failure, and it is worth reading before the aggregate: the single percentage conceals a sharp qualitative split.

On gold, ORDER and CONTROL are perfect ($45/45$ and $15/15$). Every rung of those ladders applies only meaning-preserving edits, which move no factor, so exact invariance is the correct answer and the model gives it. CONFLICT is strong at $89/112$ (79.5%). CHAIN is the weak gold ladder at $18/30$ (60.0%): its rungs differ by removing entailment links from a chain whose remaining entailments already concentrate most of the mass, the smallest perturbation the corpus contains and the one most exposed to i-bound approximation.

Every mined arm scores $0/45$ *on* ORDER *and* $0/15$ *on* CONTROL. This is the most consequential number in the table and it is not a scoring artefact. Those ladders hold the claim set fixed and permute the prose; a relation graph read off the text should therefore be identical across their rungs. It is not. On one ORDER family the `gpt-oss` WINDOWED miner emitted 9, 16, 20, 20 and 17 edges across five rungs that differ *only* in sentence order, and LCS_{cn} swung by ± 0.49 between adjacent rungs where the required answer was exact invariance. So the extractor is strongly order-dependent: the same claims, reordered, yield a materially different graph. Human annotators are the obvious reference point here and we do not have them, so we state the weaker claim the data supports — LLM relation extraction is not reorder-invariant, and any coherence measure built on it inherits that sensitivity however sound its probabilistic layer.

The corollary is that the mined arms are *not* uniformly worse than gold. On the two ladders that demand a monotone trend they are respectable — CONFLICT 53–59 of 112 and CHAIN 12–18 of 30,

Table 9: Declared ordering assertions satisfied on the ladder corpus, under two different judge models. Only the readout-independent content of a constraint is scorable by a single-score baseline: the four C1 consecutive-increase pairs and the C3 endpoint separation, giving five per family. C2 is excluded because it predicts the internal behaviour of particular readouts (§5.5), which a baseline makes no claim about. The ORDER and CONTROL families assert invariance rather than increase and so contribute *no* scorable assertion, leaving ten families and 50 assertions; the LCS rows are scored on *exactly* those 50, so they are comparable to the rows below them, while the 82.7% of Table 8 is computed over the fuller 202-assertion set including invariance and is not. Judges are the mean of five runs. Model-free baselines are identical under both judges, as they must be; the two judge rows are the entire difference.

Measure	Assertions satisfied				
	`llama-3.3-70b`		`gpt-oss-120b`		Uses LLM
LCS$_{\mathrm{mm}}$ (gold relations)		37/50	**74.0**		no
LCS$_{\mathrm{lp}}$ (gold relations)		36/50	72.0		no
LCS$_{\mathrm{cn}}$ (gold relations)		34/50	68.0		no
RC (cohesion)	30/50	60.0	30/50	60.0	no
LC (cohesion)	28/50	56.0	28/50	56.0	no
LENGTH (control)	20/50	40.0	20/50	40.0	no
ROSCOE-SC (mean)	16/50	32.0	17/50	34.0	yes
CONTRADICTION RATE	12/50	24.0	12/50	24.0	yes
JUDGE, direct rating	4/50	8.0	18/50	36.0	yes
JUDGE, G-Eval	0/50	0.0	21/50	42.0	yes
ROSCOE-SC (max)	0/50	0.0	0/50	0.0	yes
ENTITY GRAPH	0/50	0.0	0/50	0.0	no
CLAIM COUNT (control)	0/50	0.0	0/50	0.0	no

the latter matching or exceeding gold's 18/30 on `llama`'s BIDIRECTIONAL arm — and they lose the aggregate comparison almost entirely on the two invariance ladders, where they score nothing at all. A measure evaluated only on increase-type ladders would have reported mined extraction as far closer to gold than it is. That is precisely the false positive the CONTROL ladder was added to catch, and here it caught one.

Two secondary observations. Mined ladder agreement does *not* track coupling F1: `llama`'s BIDIRECTIONAL arm has the worst coupling F1 of the four (29.3) and the best ladder agreement (37.1%), while `gpt-oss`'s WINDOWED arm inverts that. Given the four-to-six assertion run-to-run movement measured in §8, the honest reading is that these four arms are not separated by ladder agreement at all, and candidate policies should not be ranked on it. And the mean-readout block shows the BIDIRECTIONAL density cost quantitatively: LCS$_{\mathrm{lp}}$ collapses to 0.037 and 0.033 against gold's 0.410 on the same seventy responses, from adding roughly two thousand spurious factors — the log-partition readout registers added factor mass most directly.

8.5 THE EVIDENCE-FREE BASELINES, MEASURED

The baselines do not all measure the same thing as the LCS, and the comparison is only readable once two constructs are separated. *Logical coherence* is a property of the relations among a response's claims: whether what it asserts can jointly hold. *Discourse cohesion* is a property of its surface texture — entity repetition, lexical chaining, referential continuity. The two are correlated in ordinary prose and come apart under exactly the edits this corpus applies, which is why the cohesion metrics appear here as baselines rather than as competitors: §7.2 gives a two-sentence pair on which all three single-document DiscoScore metrics return *identical* values while one response contradicts itself and the other does not. A cohesion metric that scores well below is measuring a different axis, not measuring this one badly.

Two results follow.

The probabilistic model separates from cohesion counting. LCS$_{\mathrm{mm}}$ reaches 37/50 (74.0%) against the best evidence-free baseline, noun-repetition cohesion, at 30/50 (60.0%). *All three* graded readouts (74.0, 72.0, 68.0) sit above *every* evidence-free baseline, so the ordering is not an artefact of choosing a headline readout, and the margin is larger than the four-to-six assertion run-to-run movement of §8. This supersedes a tie between the two that we previously reported at this comparison's smaller scale; we flag the direction of the correction rather than restating the number quietly.

An LLM judge's score on this benchmark is a property of the judge, not of the method. Every model-free baseline is bit-identical across the two judge runs — RC $30/50$ twice, LC $28/50$ twice, LENGTH $20/50$ twice — which is the internal check that the harness is deterministic where it should be. Against that fixed background the two judge rows move enormously: G-Eval scores $0/50$ under `llama-3.3-70b` and $21/50$ under `gpt-oss-120b`, and direct rating $4/50$ against $18/50$. Same prose, same rubric, same five-run averaging, same 50 assertions; only the judge model changed. A single-judge evaluation of this kind therefore reports the judge as much as the metric, and the 42-point spread is larger than the gap between the best baseline and the worst.

The mechanism is saturation rather than noise, and the two judges fail in opposite ways. `llama`'s G-Eval is astonishingly stable and useless: mean per-item standard deviation 0.0002, only three distinct values across seventy items, and a mean score of 0.996 — it rates almost everything at ceiling, so it cannot order anything. `gpt-oss`'s G-Eval uses its range (fourteen distinct values, mean 0.749) and pays in stability (mean sd 0.0800, max 0.187). Reporting judge variance alone would have called the first judge the reliable one, which is exactly backwards for the task.

Finally, four baselines score $0/50$ under both judges and it is worth separating the reasons, since they are not one finding. ROSCOE-SC (max) collapses to a constant 0.0 on all seventy items: a $\max$ over pairwise contradiction probabilities saturates as soon as one pair is confidently contradictory, which on prose of this length is every item. ENTITY GRAPH returns a constant 1.0. CLAIM COUNT is constant *by construction*: every rung of a family shares one claim set, which is what makes this corpus a controlled contrast and what makes a claim-count proxy vacuous on it. Only the first two are failures of the measure; the third is the corpus doing its job.

8.6 SUMMARY: WHAT THE LCS ADDS, AND WHAT THE BASELINES CANNOT DO

The corpus asks two questions of a coherence measure, and they are not the same question. On the CONFLICT and CHAIN families it must *move in the declared direction* as relations are repaired — 50 assertions readable by any single-score measure. On the ORDER and CONTROL families it must *not move at all*, because those rungs permute prose while holding the relation set fixed — 20 further pairs. A measure is only useful if it passes both: ordering without invariance is a length or edit-count proxy, and invariance without ordering is a constant.

What the LCS adds. Given a correct relation graph it is the only measure here that passes both axes non-vacuously: $37/50$ on ordering and $20/20$ on invariance, and the same holds for the other two graded readouts (Table 10). The invariance column is where the gap is categorical rather than incremental. The readouts are invariant under prose reordering *by construction*, not by fitting: a PRECEDENCE↔SUCCESSION swap compiles to Level-1 `none` and contributes no factor, and shuffling sentences does not change which claims stand in which relation, so the Markov network is literally identical and the readout cannot move. No surface statistic can offer that guarantee, because every surface statistic is a function of the surface. This is the property the construction buys, and it is the reason a coherence measure should be defined over relations rather than over text.

What the baselines fail to do, and it is three different failures. They are not weak in one way. *Cohesion metrics* (RC, LC) are the strongest baselines and get respectably far on ordering (60%, 56%) while being nearly flat on invariance ($16/20$, drifting by at most 0.0096 across a permuted family) — they fail not by being noisy but by measuring a different axis, so they cannot distinguish a response that contradicts itself from one that does not when both repeat the same nouns. *Contradiction-based metrics* (ROSCOE-SC, CONTRADICTION RATE) detect conflict soundly and then destroy the signal in aggregation: the $\max$ variant is a constant on prose of this length, and the mean variant divides by $\binom{n}{2}$, so both are dominated by response size rather than by relational structure. *LLM judges* are the least reliable of the three, and in a way that a single-judge evaluation conceals: G-Eval scores $0/50$ under one judge and $21/50$ under another on identical prose and rubric, and both judges violate

Table 10: Both axes at once, on the ladder corpus. **Increase** is the 50 readout-independent ordering assertions over the eight CONFLICT and two CHAIN families; **Invariance** is the 20 rung pairs over the three ORDER and one CONTROL families, satisfied when the score is unchanged to 10^{-6}. Baselines are shown under both judge models where the two differ; the model-free rows are judge-independent by construction. The three baselines at $20/20$ invariance reach it *vacuously* — each is a constant over all seventy items, hence trivially invariant and trivially unable to order anything, which is why the final column takes the weaker of the two axes.

Measure	Increase (of 50)		Invariance (of 20)		min
LCS_{mm} (gold relations)	37	**74**	20	**100**	**74**
LCS_{lp} (gold relations)	36	72	20	100	72
LCS_{cn} (gold relations)	34	68	20	100	68
RC (cohesion)	30	60	16	80	60
LC (cohesion)	28	56	16	80	56
LENGTH (control)	20	40	12	60	40
ROSCOE-SC (mean)	16–17	32–34	5–9	25–45	32
CONTRADICTION RATE	12	24	8–9	40–45	24
JUDGE, G-Eval	0–21	0–42	5–13	25–65	0
JUDGE, direct rating	4–18	8–36	5–14	25–70	8
ROSCOE-SC (max)	0	0	20	100[†]	0
ENTITY GRAPH	0	0	20	100[†]	0
CLAIM COUNT (control)	0	0	20	100[†]	0

[†] vacuous: the measure is constant across all seventy items.

invariance freely — `gpt-oss`'s G-Eval moves by up to 0.55 across rungs that differ only in sentence order. A judge asked for a coherence rating is answering a question about the text, so it inherits the text's surface variation.

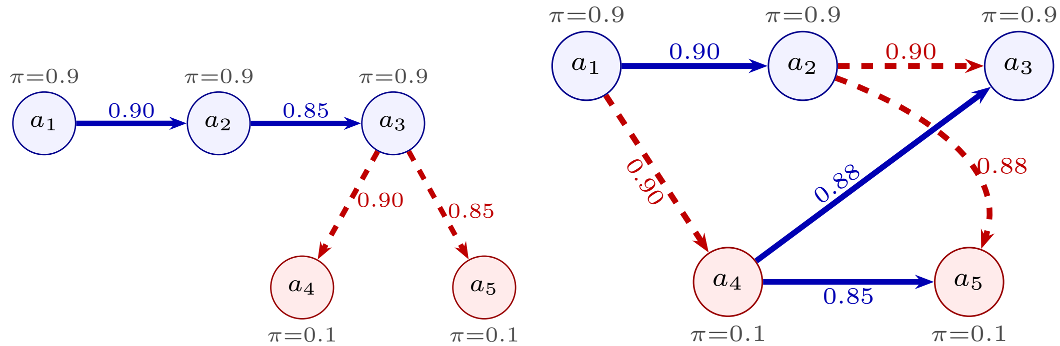
Two honest qualifications. First, the LCS column above is the *gold-relation* arm. Handed graphs mined from prose the same readouts fall to 34.7–37.1% of the full 202 assertions and to $0/45$ and $0/15$ on the invariance ladders (§8.5, Table 8) — the guarantee is a property of the model given a correct graph, and mined graphs are not yet reorder-invariant. The contribution is therefore a measure that is right when its input is right, plus a benchmark that localizes the remaining problem in extraction rather than in scoring. Second, the two axes are not equally hard: ten of the thirteen rows clear 40% on invariance while only five clear 50% on ordering, and of those five, three are the LCS readouts and two are the cohesion metrics. Invariance alone is therefore not a discriminating test — it is a necessary condition most baselines meet, and the three constant baselines meet trivially. The claim rests on passing both, which only the LCS does.

9 WORKED EXAMPLES

We work four examples. The first is the smallest complete instance of the two-stage model: one claim set, two responses, exact inference. The second is a richer sixteen-claim graph that exercises the conflict taxonomy. The third is a claim-identical pair that isolates ordering. The first three are hand-authored, which bounds what they can show: they demonstrate that the model reads a graph correctly, not that the estimator finds such graphs in text it has never seen. The fourth is therefore a corpus rather than an example — one hundred machine-generated responses whose incoherence is planted by construction, scored end to end with no hand-drawn relations anywhere.

9.1 A FIVE-CLAIM PAIR, END TO END

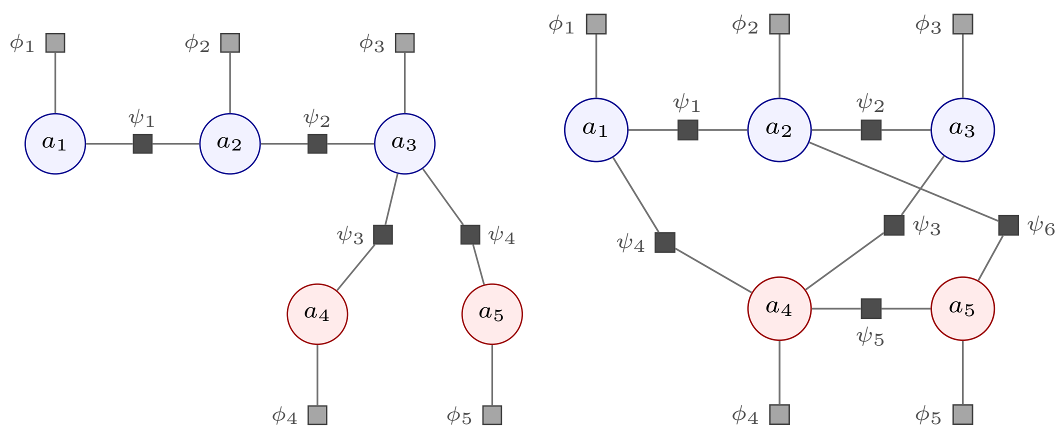
The point of this example is that factuality and coherence are separable, so we hold the first fixed and vary only the second. Five claims about the Voyager 1 mission are fixed once and for all, three of them true of the world and two false; stage one is taken as given, contributing the priors $\pi = 0.9$ for each true claim and $\pi = 0.1$ for each false one through Eq. 9:



(a) **Response A, coherent.** The three true claims form a support chain ($a_1 \to a_2 \to a_3$) and each false claim is *attributed and rejected* by a CONTRADICTION edge from a_3. Every claim ends on the correct side of its own prior: $\mathrm{LCS}_{\mathrm{mm}} = 0.590$, $\mathrm{LCS}_{\mathrm{cn}} = 0.963$, $\log Z = -0.95$.

(b) **Response B, incoherent.** The same five claims. The response contradicts its own chain ($a_2 \not\to a_3$) and makes the false a_4 a load-bearing premise for the true a_3, while a_1 and a_2 deny a_4 and a_5. The *true* claim a_3 is dragged to 0.513 — far below its own 0.9 prior: $\mathrm{LCS}_{\mathrm{mm}} = 0.494$, $\mathrm{LCS}_{\mathrm{cn}} = 0.414$, $\log Z = -3.78$.

Figure 2: Relation graphs for the five-claim worked pair (§9). Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); the *same* five claims appear in both panels. Solid blue: ENTAILMENT; dashed red: CONTRADICTION. Line width grows with the relation weight p. The panels differ only in *how the response arranges* the claims, which is the whole of what coherence measures — no per-claim factuality check can separate them.



(a) **Response A.** Five unaries and four pairwise factors.

(b) **Response B.** The same five unaries, but six pairwise factors wired so that ψ_3 makes a false claim a premise for a true one.

Figure 3: The coherence MRF of Eq. 2 for the two responses, as an explicit factor graph. Circles are the binary claim variables; grey squares the unary factors $\phi_i = [1 - \pi_i, \pi_i]$ that carry the priors; black squares the pairwise factors ψ_{τ_r, p_r} of Table 2, one per mined relation. **The priors enter only through the** ϕ_i: the claim-level potentials are identical across the two panels, and every difference in score comes from the ψ_r — their number, their endpoints, and their couplings.

		π_i	Claim
a_1	true	0.9	Voyager 1 crossed the heliopause in August 2012.
a_2	true	0.9	After the crossing, Voyager 1's instruments measured a sharp rise in plasma density.
a_3	true	0.9	Voyager 1 is therefore in interstellar space.
a_4	false	0.1	Voyager 1's plasma wave instrument failed permanently before the crossing.
a_5	false	0.1	Voyager 1 has left the Solar System entirely.

Two responses then assert *these same five claims* and differ only in how they arrange them. Because the claim set is identical, every per-claim factuality check returns the same verdict on both; the

Response A (coherent) — $\text{LCS}_{\text{mm}} = 0.590$, $\text{LCS}_{\text{cn}} = 0.963$, $\log Z = -0.95$
Voyager 1 crossed the heliopause in August 2012. $[a_1]$ After the crossing, its instruments measured a sharp rise in plasma density, $[a_2]$ and that measurement is what established the boundary had been passed; Voyager 1 is therefore in interstellar space. $[a_3]$ Two claims that circulate about the mission do not survive scrutiny. It is sometimes said that the plasma wave instrument had failed permanently before the crossing, $[a_4]$ *but it cannot have failed and also have returned the density measurement that settled the question.* It is also said that the probe has left the Solar System entirely; $[a_5]$ *being in interstellar space is not the same thing, since the Oort cloud remains gravitationally bound and lies far beyond Voyager 1.*
Response B (incoherent) — $\text{LCS}_{\text{mm}} = 0.494$, $\text{LCS}_{\text{cn}} = 0.414$, $\log Z = -3.78$
Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and after the crossing its instruments measured a sharp rise in plasma density. $[a_2]$ *That said, the probe cannot really be counted as being in interstellar space.* $[a_3]$ The plasma wave instrument had in fact failed permanently before the crossing, $[a_4]$ *and it is precisely that failure which tells us the probe reached interstellar space.* The instrument failure also means Voyager 1 has left the Solar System entirely. $[a_5]$ *Of course, the August 2012 crossing rules out any such permanent failure, and the density measurement is flatly inconsistent with the probe having left the Solar System.*

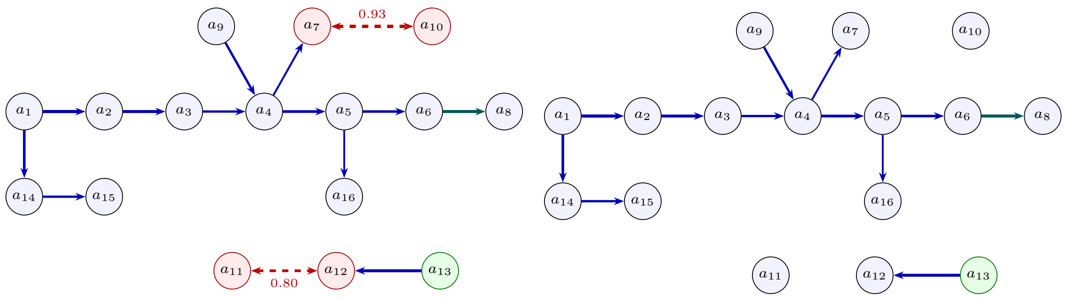
Table 11: The two responses in full. They assert the *same* five claims of §9.1; each claim's first mention is marked $[a_i]$. Italicized spans are the ones the estimator reads as relations. In A they discharge the two false claims; in B they reason *from* them.

relation graphs (Figure 2) and the MRFs they induce (Figure 3) are what differ. Table 11 gives both responses in full, with the span asserting each claim marked $[a_i]$ at its first mention.

Table 12: The mined relations of both responses, one row per claim pair. S is the Level-2 discourse sense, τ the Level-1 coupling it compiles to (Table 3), p the weight. E abbreviates ENTAILMENT, C CONTRADICTION. These are exactly the edges of Figure 2 and the factors ψ_r of Figure 3; a dash means the response draws no relation over that pair.

	Response A (coherent)				**Response B** (incoherent)				
Pair	Edge	S	τ	p	Edge	S	τ	p	What differs
$\{a_1, a_2\}$	$a_1 \to a_2$	Cau-Eff	E	.90	$a_1 \to a_2$	Cau-Eff	E	.90	The one relation both agree on.
$\{a_2, a_3\}$	$a_2 \to a_3$	Evidence	E	.85	$a_2 \to a_3$	Contrast	C	.90	A supports its conclusion; B denies it. Support becomes conflict.
$\{a_3, a_4\}$	$a_3 \to a_4$	Contrast	C	.90	$a_4 \to a_3$	Evidence	E	.88	**The arrow reverses.** A rejects a_4; B makes it a premise.
$\{a_3, a_5\}$	$a_3 \to a_5$	Contrast	C	.85		—			A refuses a_5; B never draws the distinction.
$\{a_4, a_5\}$		—			$a_4 \to a_5$	Cau-Eff	E	.85	B alone chains one falsehood into the other.
$\{a_1, a_4\}$		—			$a_1 \to a_4$	Contrast	C	.90	B denies a_4, which it had just used as a premise — this strands a_3.
$\{a_2, a_5\}$		—			$a_2 \to a_5$	Contrast	C	.88	Likewise for a_5: asserted, then denied.
Totals		2 E, 2 C				3 E, 3 C			A's conflicts point *outward*, at the false claims; half of B's point *inward*.

The italicized spans are the ones the estimator reads as relations, and they are the whole difference between the two. Table 12 lays the two relation sets side by side, one row per claim pair, with each relation's discourse sense and the coupling it compiles to. In A the false claims are named, attributed, and then refused on a stated ground, which compiles to a CONTRADICTION edge from a_3. In B the same falsehoods are named and then *used* — a_4 is offered as the reason a_3 holds — while the response also denies a_3 outright and, in its last sentence, denies a_4 and a_5 too. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. All values below are exact, by $2^5 = 32$-world enumeration.



(a) **Incoherent** (base). Two EXCLUSIVE conflicts (dashed): $a_7 \leftrightarrow a_{10}$ is *unresolved*; $a_{11} \leftrightarrow a_{12}$ is a concession the holding a_{13} (green) resolves. $\mathrm{mm} = 0.607$, $\log Z = -9.64$.

(b) **Coherent**. The identical support structure with both conflict edges removed; a_{10} and a_7 are no longer suppressed. $\mathrm{mm} = 0.620$, $\log Z = -8.25$, $\mathrm{lp} = 1.00$.

Figure 4: Relation graphs for the AeroParts recall report (§9). Solid blue: ENTAILMENT; teal: EQUIVALENCE; dashed red, double-headed: EXCLUSIVE. Line width grows with the relation weight p. **The two panels differ only in the two conflict edges**, so the $0.607 \to 0.620$ lift and the 1.39 nats of recovered mass are attributable to conflict structure alone.

Example 2 (Coherent)**.** *Response A's four relations (Table 12, column A) are a two-edge support chain over the true claims and one rejection of each false claim. The model returns* $\mathrm{LCS}_{\mathrm{mm}} = 0.590$ *and* $\mathrm{LCS}_{\mathrm{cn}} = 0.963$ *with* $\log Z = -0.95$*. Every claim finishes on the correct side of its own prior: the true claims sit at* 0.939*,* 0.989 *and* 0.992 *— each* above *its* 0.9 *prior, because the chain supplies mutual support — and the false ones at* 0.013 *and* 0.020*, pushed below* 0.1 *by the rejections. Nothing in the response fights anything else, so no claim is penalized.*

Example 3 (Incoherent)**.** *Response B keeps only* $a_1 \to a_2$ *and replaces the rest with six relations that pull against each other (Table 12, column B): it denies its own conclusion, promotes the false* a_4 *to a premise for that same conclusion, chains* a_4 *into* a_5*, and then denies both falsehoods in its closing sentence. Three of its six relations are conflicts, and two of those point at claims the response itself asserts. The score falls to* $\mathrm{LCS}_{\mathrm{mm}} = 0.494$ *and* $\mathrm{LCS}_{\mathrm{cn}} = 0.414$*, with* $\log Z = -3.78$*. The diagnostic that localizes the damage is per-claim: the* true a_3 *is dragged to* 0.513*, far below its own* 0.9 *prior, because its only support arrives from* a_4*, which the response itself denies — and a premise the text refuses cannot transmit support. Three claims finish below their own priors, against two in Response A, and* a_3 *is the one that matters: a claim true of the world, demoted to a coin flip by the argument built around it.*

Two things are worth reading off this pair. First, the priors enter *only* through the unary factors ϕ_i of Figure 3, which are identical in both panels; the entire $0.590 \to 0.494$ drop is produced by the ψ_r, so it is attributable to arrangement alone. Second, a claim being true of the world does not protect it: coherence is a property of the joint distribution, and a_3 is suppressed by *how* it is argued for, not by what it says. This is the mechanism Remark 2 anticipates and the ordering pair of §9.3 exhibits at scale.

A caution on readouts. $\mathrm{LCS}_{\mathrm{lp}}$ is 0.194 for Response A and 0.267 for Response B, which inverts the order — correctly, and for a reason Eq. 8 makes plain: its references $\log Z_{\max}$ and $\log Z_{\min}$ are recomputed *per response* (-0.49 and -1.06 for A, -1.65 and -4.55 for B), so it reports how close a response comes to the ceiling of *its own* edge skeleton, not how the two responses compare. It is a within-response diagnostic. Raw $\log Z$ is comparable across responses and moves the right way, by 2.83 nats.

9.2 A SIXTEEN-CLAIM RECALL REPORT

The second example is a sixteen-claim aviation recall report, which exercises the conflict taxonomy more fully than the five-claim pair: it carries both an unresolved conflict and a conceded one. Its relation graph is twelve support edges — an entailment spine $a_1 \to a_2 \to \ldots \to a_6$ (shipment certified to spec, installation, entry into service, in-flight failures, regulatory probe, grounding order), branches to the pulled-from-service claim, the maintenance-log evidence, the tribunal's damages award and the regulator's bulletin, and one EQUIVALENCE edge — plus two conflicts. Every value below is exact, by 2^{16} enumeration, and the graph is reproduced edge-by-edge in Figure 4 with each edge's weight shown by line thickness.

Example 4 (Incoherent)**.** *The report asserts both that* no passengers or crew were harmed *and that a wire report* claimed three passengers died*. These are exhaustive alternatives rather than mere opposition: they cannot both hold and cannot both fail, so the relation is* EXCLUSIVE *with* $p = 0.93$ *— and nothing in the response resolves it. A second tension, between the airline's pilot-error argument and the final report's metallurgical-defect finding (*$p = 0.80$*), is a* Concession *that the tribunal's holding does resolve, and Eq. 3 discounts it accordingly. The model returns* $\mathrm{LCS}_{\mathrm{mm}} = 0.607$ *with* $\log Z = -9.64$*. The diagnostic that localizes the incoherence is per-claim: the marginal of the claim on the losing side of the* unresolved *conflict is dragged* below its own prior*, which no aggregate score would reveal. Applying the concession discount alone — resolving the second tension while leaving the first — raises the score to* 0.612*, as C2 requires.*

Example 5 (Coherent)**.** *Removing the two conflict edges and changing nothing else leaves the identical support structure (Figure 4b). The score rises to* $\mathrm{LCS}_{\mathrm{mm}} = 0.620$ *with* $\log Z = -8.25$ *and* $\mathrm{LCS}_{\mathrm{lp}} = 1.00$*: the response is now as coherent as its own edge skeleton permits, which is what the ceiling of Eq. 8 means. The suppressed claim recovers to its prior.*

Because the two panels of Figure 4 differ *only* in the two conflict edges, the $0.607 \to 0.620$ lift and the 1.39 nats of recovered probability mass are attributable to conflict structure alone — which is the property the ladder construction of §7 generalizes from one pair to a family. The full ladder over this family is monotone throughout: $0.586 < 0.607 < 0.612 < 0.613 < 0.620$ over the rungs *worse*, *base*, *concession-resolved*, *one-conflict-fixed*, *coherent*.[1]

9.3 A CLAIM-IDENTICAL ORDERING PAIR

This pair makes the order-sensitivity claim concretely. Two summaries of one appellate judgment are built from an *identical* set of eighteen claims — the same atom texts, one a permutation of the other — differing only in whether self-serving assertions precede or follow the findings that qualify them. The faithful ordering scores $\mathrm{LCS}_{\mathrm{mm}} = 0.95$ and the self-serving-first ordering 0.90. No per-claim factuality check can separate them, because every claim is byte-identical; what differs is which claims the response leaves standing unqualified at the point where it asserts them.

It is worth being exact about *how* the two scores differ, because Remark 2 might seem to forbid it: nothing in Eq. 2 references position, so an edit preserving the relation set cannot move any readout. The pair does not preserve the relation set. Mining is response-grounded (§4.5), so the relations the estimator draws are a function of the prose, not of the claim list: the faithful summary states each tension beside the truth or holding that settles it, which the estimator reads as a Concession the text resolves and Eq. 3 discounts; the adversarial summary leaves the same claims standing unqualified until a later holding overrides them, which reads as a live conflict. The claim set is what factuality sees and it is identical; the relation set is what coherence sees and it is not.

10 LIMITATIONS

The model is pairwise, and some incoherence is not. Definition 1 admits only pairwise factors, so genuinely k-ary structure lies outside the family. This is the sharpest limitation of the measure and it is worth stating exactly rather than in passing, because it is not a matter of degree: the relevant conflicts are not approximated badly, they are unrepresentable.

Proposition 4 (Jointly inconsistent, pairwise consistent)**.** *Let* $\mathbb{P}(\mathbf{a})$ *factorize as in Eq. 2 over* a_1, a_2, a_3 *with non-negative factors. If* $\mathbb{P}(a_1{=}1, a_2{=}1, a_3{=}1) = 0$ *then* $\mathbb{P}(a_i{=}1, a_j{=}1) = 0$ *for some pair* $i \neq j$*. Consequently no member of the family represents "the three claims cannot all hold, but any two can".*

[1] Treating both tensions as plain CONTRADICTION rather than EXCLUSIVE — the three-coupling model that omits the exhaustiveness distinction — gives $\mathrm{LCS}_{\mathrm{mm}} = 0.587$ and $\log Z = -9.75$ for Example 4, with Example 5 unchanged at 0.620 and -8.25. The distinction of §4.2 is worth 0.020 of score here, and the reason it matters more than that figure suggests is Table 5: with EXCLUSIVE conflicts, readout (b)'s conflict term is pinned at 1.000 and the readout fails outright.

Proof. By Eq. 2, $\mathbb{P}(1,1,1) = Z^{-1} \prod_i \phi_i(1) \prod_{i<j} \psi_{ij}(1,1)$, a product of non-negative terms, which vanishes only if some term does. If $\phi_i(1) = 0$ then a_i is false almost surely and both pairs containing i have probability zero. If $\psi_{ij}(1,1) = 0$ then every world with $a_i = a_j = 1$ carries that factor, so $\mathbb{P}(a_i{=}1, a_j{=}1) = 0$ directly. A ternary factor that vanishes only at $(1,1,1)$ has neither consequence. $\square$

The gap is quantitative as well as qualitative. With uniform priors, a ternary factor supported off $(1,1,1)$ gives $\mathbb{P}(1,1,1) = 0$ while leaving each pair jointly true with probability 0.143; the best pairwise attempt that drives $\mathbb{P}(1,1,1)$ below 10^{-4} leaves the pairs at 0.028, and a random search over 2×10^5 pairwise parameterizations found nothing better. The pairwise model can only suppress the bad world by also suppressing the good ones that share a $(1,1)$ cell with it. Every figure in this paragraph is exact by 2^3 enumeration and reproduced by a released script and regression test, so the proposition is checked and not merely asserted.

This is not a hypothetical. Prompting a frontier model to plant a single arithmetic inconsistency in an otherwise ordinary encyclopedic paragraph produced a passage whose decomposition places the conflict across *three* claims: that a league comprises 356 teams, that those teams form 32 conferences, and that each conference has exactly 10 member schools. No two of the three conflict; $32 \times 10 \neq 356$ requires all three. Mining that passage with the estimator of §4.5 returns *zero* conflict edges, and the readouts accordingly report it as coherent ($\mathrm{LCS}_{\mathrm{mm}} = 0.68$, $\mathrm{LCS}_{\mathrm{lp}} = 1.00$). The omission is not an extraction failure: run under the `all_pairs` policy, which scores every one of the $\binom{n}{2}$ candidate pairs, the result is unchanged, and querying the estimator on each of the three pairs in isolation returns NONE or ENTAILMENT — never a conflict. Proposition 4 says no reweighting of those factors could have helped.

Two milder consequences of the same restriction are worth naming. Transitivity is not enforced — the model will happily hold $a \Rightarrow b$ and $b \Rightarrow c$ at high weight while leaving a and c unrelated, since it has no rule tying them. And a claim conceded by two independent holdings is discounted twice rather than once. All three are expressible as first-order rules in the Markov logic view of Appendix C, which is the natural route past this limitation and which we have not taken; Remark 1's closure guarantee is precisely what buys tractability here and precisely what costs us this class of defect.

Hand-set constants. The design deliberately avoids free parameters — criterion R4 exists for that reason, and it is what disqualifies readout (c) with its prior ρ. Two constants remain. The concession discount $\lambda = 0.45$ in Eq. 3 sets how much a resolved tension is softened, and the resolver-detection heuristic identifies holdings by cue phrases and lexical overlap. Both are defensible and neither is derived. A principled treatment would learn the discount from labelled resolved and unresolved concessions, which the corpus of §7 supplies in principle but not at the scale required.

Relation estimation dominates the error. §8 is unambiguous that the bottleneck is not the model. At 56.3% coupling-level F1, roughly half the factors in a mined network are wrong or missing, and the 25–43 point ladder-agreement gap between mined and gold arms follows from that. Everything the paper establishes about the readouts is established on gold or exact graphs; the mined-arm numbers should be read as a measurement of current extraction quality, not as the measure's ceiling.

Approximate inference is not characterized. Production inference is weighted mini-bucket with i-bound 6, while every exact value reported here comes from 2^n enumeration, feasible only for $n \leq 20$ claims. The two agree on the fixtures we can check both ways, but we have not characterized the approximation error at length, and the augmented networks of §5.2 — which add one auxiliary variable per relation — are the case where it would most plausibly bite.

Empirical scope. The generated corpus is fourteen families and seventy items spanning all four ladder types, with single-sample mining whose measured run-to-run movement is at most 0.014 F1 across two independent runs of the same configuration. All five baseline families of §7.1 are measured on the ladder corpus (§8.5). On the 50 readout-independent ladder assertions all three graded readouts now sit above every evidence-free baseline, which supersedes the tie we reported on ten assertions; we flag the direction of that correction in §8.5 rather than quietly restating it. Liu & Strube (2025) is cited rather than run, for the output-type reason given above.

Three things remain genuinely open. First, the control and order ladders — the invariance experiments — reveal that *mined* graphs are strongly order-dependent ($0/45$ and $0/15$ on every mined arm, with a five-rung edge count of $9, 16, 20, 20, 17$ under pure sentence reordering), and we cannot yet say how a human annotator would score those same rungs, which is the comparison that would separate "the extractor is unstable" from "the invariance requirement is too strict". Second, human correlation generally, which is what would establish that the construct being ordered is the one readers care about. Third, judge dependence: the two LLM judges we ran disagree by up to 42 points on the identical assertions, so any single-judge number here — including ours — carries that uncertainty. And the corpus is LLM-generated throughout, gold relation labels included, with LLM-committee validation; a bias shared between generator and committee would not be caught by it.

11 Conclusion

Coherence is a property of a joint distribution over what a response asserts, and it can be read off a Markov random field whose factors are the relations the response itself draws. Three things follow from taking that view seriously. First, the factor tables matter: the ones that are right for grounding a claim against evidence charge a premise for being a premise, and a coherence measure built on them spends its dynamic range in the wrong place (Prop. 1). Second, the relation vocabulary can be rich at the annotation layer and must be narrow at the inferential one; twelve discourse senses compiling deterministically onto five couplings, each fixed by the world it penalizes, keeps the model a closed pairwise family while leaving the labels interpretable. Third, and least expected, not every reading of "how coherent is this?" is admissible. Of four candidates, two are non-monotone in conflict resolution for a single diagnosable reason — they measure whether a conflict is active rather than what should be believed (Prop. 3) — and a fifth, apparently the most natural of all, is maximized by asserting no relations whatever (Prop. 2). The mean posterior marginal survives all five criteria, needs one inference query, and localizes an incoherence to the claims responsible.

Measuring such a score requires controlled contrasts, and ladders of responses over one claim set with declared orderings supply them. Given gold relation graphs the readouts recover 82.7% of those orderings; given graphs mined from the response text, 34.7–37.1%. Because the two conditions differ in nothing but the relation graph, the gap localizes the open problem: the probabilistic model is adequate, and grounded relation extraction, at 52% coupling-level F1 at best and not invariant to sentence order, is the bottleneck. That is a more tractable target than "measure coherence", and it is one that improvements can be measured against.

AI use statement

Generative AI is both an object of study and a component of the artifacts released here, and we disclose both roles. The relation estimator of §4.5 is LLM-based by design: its prompts, its logprob-derived confidences and its failure modes are contributions of the paper. The evaluation corpus of §7 is LLM-generated throughout, including its gold relation labels, and validated by an independent LLM committee; §10 states the consequences for treating it as ground truth. We additionally used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that pins the exact numerical values reported in §9 and Table 5; every number in §8 was traced to a stored experiment record. The research questions, the taxonomy, the readout definitions, the propositions and their proofs are the authors' own. We take responsibility for the final content of this work.

Ethics statement

This work involves no human subjects and no personally identifying data. The fixtures include a summary of a published appellate judgment, used as a worked example of ordering-dependent coherence; it is a matter of public record and no private individual is the subject of analysis. The principal ethical risk is misreading: a coherence score is not a truth score, and a response can be perfectly coherent and entirely false. That is precisely why §6 keeps the two axes separable, and we caution against deploying LCS alone as a factuality filter — a fluent, internally consistent fabrication is exactly the artifact it scores highest. The released fixtures and corpus deliberately contain false claims, labelled as such, and should not be redistributed without those labels.

Reproducibility statement

The model is fully specified in the paper: Definition 1 gives the joint, Table 2 both families of potentials, Def. 2 the penalized sets, Table 3 the compile map, and Eqs. 5–8 the four readouts together with the auxiliary-variable constructions two of them require. Algorithms 1 and 2 give candidate selection and the end-to-end procedure. Appendix A reproduces all four estimation prompts verbatim, including the trailing token boundary that the surrogate readout depends on. The exact values in §9 and Table 5 are computed by explicit 2^n enumeration — 2^5 for the five-claim pair, 2^{16} for the recall report — and are pinned by regression tests in the released code, so they are reproducible without LLM access. The relation graphs they use are given edge-by-edge in Figures 2 and 4, the five-claim pair's claim set, priors and gold relations ship as a fixture, and a single script recomputes every number it reports. The k-ary representability figures of §10 (Prop. 4) are likewise exact by 2^3 enumeration, with the seeded random search and the published table both reproduced by a released script and pinned by regression tests. §8 names every model used and states the sampling regime and its measured variance. Code, fixtures and corpus are released.

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A THE ESTIMATION PROMPTS, VERBATIM

The four prompts below are reproduced exactly as they are issued, extracted programmatically from the released source so that this appendix cannot drift from the code. Placeholders in double braces are substituted at call time; `{{sense_menu}}` is substituted at build time from one of the two menus in §A.5.

A.1 PROMPT A, FIRST GENERATION — DISCOURSE SENSE AND COUPLING

6,148 characters. Placeholders: `{{sense_menu}}`, `{{response}}`, `{{atom_a}}`, `{{atom_b}}`.

```


Instructions:
You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
↪  taken FROM THAT RESPONSE, in their order of appearance (A comes before B). Your task is
↪  to decide the discourse/logical relation FROM A TO B, following the steps below.

IMPORTANT -- ground your decision in the response. Assert a coupling ONLY if the response
↪  ITSELF draws that connection between A and B (as written, or as a clear step in the
↪  author's argument/narrative). Do NOT assert a relation that is merely plausible in
↪  general but that the response does not actually make. If A and B both appear in the
↪  response yet the response draws no logical or discourse dependence between them, the
↪  answer is None.
```

```

1. Reason step by step, referring to the response: does the response present A as causing, enabling, providing evidence for, restating, elaborating, temporally preceding, contrasting with, or contradicting B? Is B a claim the response later withdraws or that a holding resolves? Consider the direction (A to B). If the response links A and B only indirectly through other claims, or not at all, that is None.

2. Name the DISCOURSE SENSE, one of: {{sense_menu}}.
   - Cause-Effect: A causes/leads to B. Effect-Cause: A is the effect, B its cause.
   - Evidence: A provides evidence for B. Condition: A is a condition for B.
   - Restatement: A and B assert the same thing. Instantiation: A is a general claim, B a specific instance (or vice versa).
   - Contrast: A and B are in opposition but NOT exhaustive (they need not cover all possibilities; both could conceivably be false).
   - Concession: A and B are in tension but the text concedes/resolves it ("although A, still B", or a holding settles it).
   - Alternative: A and B are EXHAUSTIVE competing options -- EXACTLY ONE holds (they are mutually exclusive AND together cover the possibilities: not both, and not neither). E.g. "no one was harmed" vs "three people died"; "the cause was pilot error" vs "the cause was a metallurgical defect".
   - Disjunction: AT LEAST ONE of A and B holds (they may both hold, but the response rules out neither being true) -- e.g. two supporting findings at least one of which must be present.
   - Precedence/Succession: A and B are ordered in time with no truth dependence.
   - None: the response draws no logical or discourse dependence between A and B.

3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive, co_necessity, none.
   - Cause-Effect, Effect-Cause, Evidence, Condition, Instantiation -> entailment
   - Restatement -> equivalence
   - Contrast, Concession -> contradiction
   - Alternative -> exclusive       (exactly one of A, B is true)
   - Disjunction -> co_necessity     (at least one of A, B is true)
   - Precedence, Succession, None -> none
   Prefer "exclusive" over "contradiction" when the two claims are not just incompatible but EXHAUSTIVE (one of them must be true); prefer "contradiction" when they merely cannot both hold but could both be false.

4. Give your final answer as two bracketed tags on ONE line, sense first:
[sense=Cause-Effect] [coupling=entailment]
A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.

Use the following examples to better understand your task.

Example 1 (the response makes the causal link):
RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns spiked, and the company's stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
1. Reasoning: the response presents the flawed launch as the head of a chain (panning, returns) that ends in the stock decline, so the response itself draws a causal link from A to B.
2. Discourse sense: Cause-Effect.
3. Coupling: A causing B is a positive inferential link, i.e. entailment.
4. Final answer:
[sense=Cause-Effect] [coupling=entailment]

Example 2 (both claims present, but the response draws NO connection -> None):
RESPONSE: The quarterly report was published in April. Separately, the annual audit was scheduled for December. The two processes are run by different teams and were not related this year.
A: The quarterly report was published in April.
B: The annual audit was scheduled for December.
1. Reasoning: both claims appear in the response, and one might imagine a reporting-to-audit link in general, but this response explicitly treats them as separate and unrelated. The response draws no dependence from A to B.
2. Discourse sense: None.
3. Coupling: no dependence the response asserts, i.e. none.
4. Final answer:
[sense=None] [coupling=none]

Example 3 (the response states an EXHAUSTIVE alternative -> exclusive):
RESPONSE: The official statement said no one was harmed in the incident. However, the coroner's report confirmed that three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
1. Reasoning: the response sets A and B against each other ("However, ...") and they cannot both be true; but they also cannot both be false -- either people were harmed or they were not -- so exactly one holds. This is exhaustive, not a mere contrast. No holding resolves it.
2. Discourse sense: Alternative.
```

```
3. Coupling: exactly one of A, B is true, i.e. exclusive.
4. Final answer:
[sense=Alternative] [coupling=exclusive]

Example 4 (at least one must hold -> co_necessity):
RESPONSE: The defect was caught in review: at least one of the two independent checks -- the vibration analysis or the metallurgical assay -- flagged it.
A: The vibration analysis flagged the defect.
B: The metallurgical assay flagged the defect.
1. Reasoning: the response asserts the defect WAS caught by at least one check, so A and B cannot both be false; but both could hold (both checks may have flagged it). This is a disjunction, not an exclusion.
2. Discourse sense: Disjunction.
3. Coupling: at least one of A, B is true, i.e. co_necessity.
4. Final answer:
[sense=Disjunction] [coupling=co_necessity]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
```

A.2 PROMPT A, SECOND GENERATION — REBALANCED FOR PRECISION

6,361 characters. Same placeholders. The five changes relative to the first generation, and their motivation, are given in §4.5: few-shot balance from 3:1 related to 2:3; negative exemplars showing silence rather than explicit disavowal; a verifiable evidence span; the discriminating question asked first; and NONE given first position and equal prose. Transitivity is named as a non-relation.

```


Instructions:
You are given the full RESPONSE a model produced, and two atomic claims, A and B, both taken FROM THAT RESPONSE. Decide the discourse/logical relation FROM A TO B.

Most pairs of claims in a text are NOT related. They are simply near each other. Your default answer is None, and you should depart from it only when the response itself draws a dependence between A and B.

Three things that are NOT relations, however plausible they look:
  - Topical adjacency. A and B being about the same subject, or sitting in neighbouring sentences, is not a relation.
  - World knowledge. If the link is true in general but the response does not draw it, the answer is None.
  - Transitivity. If the response links A to some third claim C, and C to B, that does NOT make A related to B. Report only the links the text draws directly.

1. First ask the discriminating question: does the response ITSELF assert a dependence from A to B -- as written, or as an explicit step in the author's argument? Identify the words in the response that do it. If you cannot point to such words, the answer is None.

2. If and only if you can, name the DISCOURSE SENSE, one of: {{sense_menu}}.
    - None: the response draws no dependence between A and B. This is the common case: unrelated, merely adjacent, related only by world knowledge, or related only through a third claim.
    - Cause-Effect: the response says A causes/leads to B. Effect-Cause: A is the effect, B its cause.
    - Evidence: the response offers A as evidence/support for B.
    - Restatement: A and B say the same thing -- a paraphrase, a converse, or a restatement the text marks ("in other words", "equivalently", "that is").
    - Contrast: A and B are opposed but NOT exhaustive (both could be false).
    - Concession: A and B are in tension and the text concedes or resolves it ("although A, still B"; a holding settles it).
    - Alternative: A and B are EXHAUSTIVE competing options -- exactly one holds (not both, not neither).
    - Disjunction: AT LEAST ONE of A and B holds; both may.
    - Precedence: A and B are ordered in time with no truth dependence.

3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive, co_necessity, none.
    - Cause-Effect, Effect-Cause, Evidence -> entailment
    - Restatement -> equivalence
    - Contrast, Concession -> contradiction
    - Alternative -> exclusive       (exactly one of A, B is true)
    - Disjunction -> co_necessity    (at least one of A, B is true)
    - Precedence, None -> none
```

```
   Prefer "exclusive" when the two claims are incompatible AND exhaustive; "contradiction" when they merely cannot both hold but could both be false.

4. Answer on ONE line, with three bracketed tags. The evidence must be a SHORT VERBATIM QUOTE COPIED FROM THE RESPONSE -- the words that draw the link. Do not quote claim A or claim B; quote the response. Use [evidence=none] when the sense is None.
[sense=Evidence] [coupling=entailment] [evidence="which is why the panel held"]

Use the following examples to better understand your task.

Example 1 (adjacent and on-topic, but the response draws no link -> None):
RESPONSE: The survey was administered in March to a sample of 400 households. Response rates in rural districts have declined over the past decade. The analysis weighted results by district population.
A: The survey was administered in March to a sample of 400 households.
B: Response rates in rural districts have declined over the past decade.
1. The two sentences are adjacent and both concern the survey, but the response never says the March administration bears on rural response rates. There are no words that draw a dependence.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Example 2 (the response makes the causal link explicitly):
RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns spiked, and as a direct result the company's stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
1. The response presents the launch as the head of a chain ending in the stock decline, and marks it: "as a direct result".
2. Discourse sense: Cause-Effect.
3. Coupling: entailment.
4. [sense=Cause-Effect] [coupling=entailment] [evidence="as a direct result"]

Example 3 (true in general, but this response does not draw it -> None):
RESPONSE: Higher interest rates raise the cost of corporate borrowing. The firm opened three distribution centres last year. Its logistics costs fell by eight percent.
A: Higher interest rates raise the cost of corporate borrowing.
B: The firm opened three distribution centres last year.
1. One could argue borrowing costs affect expansion decisions, but that is world knowledge. This response states the two facts side by side and links neither to the other.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Example 4 (the response states an exhaustive alternative -> exclusive):
RESPONSE: The two readings are mutually exclusive and exactly one must be right: either no one was harmed in the incident, or three people died in it.
A: No one was harmed in the incident.
B: Three people died in the incident.
1. The response sets A and B against each other and says exactly one must be right: "mutually exclusive and exactly one must be right: either ... or".
2. Discourse sense: Alternative.
3. Coupling: exclusive.
4. [sense=Alternative] [coupling=exclusive] [evidence="mutually exclusive and exactly one must be right"]

Example 5 (linked only through a third claim -> None):
RESPONSE: The alloy is chemically identical to the certified reference. It therefore meets the reference specification. Meeting that specification is a precondition for airframe use.
A: The alloy is chemically identical to the certified reference.
B: Meeting that specification is a precondition for airframe use.
1. The response links A to the middle claim (it meets the specification), and the middle claim to B. It does not link A to B directly; that would be transitivity, which is not a relation to report.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
```

A.3 PROMPT B, DEFAULT — SURROGATE-TOKEN CONDITIONAL STRENGTH

2,182 characters. Placeholders: `{{response}}`, `{{atom_a}}`, `{{atom_b}}`, `{{coupling}}` (twice). This is the one prompt that does not end in a newline: it terminates with `Answer: ` including the trailing space, so that the model's next token is the surrogate whose logprobs are read. Reproducing it without that trailing space changes the tokenization of the answer position and therefore the estimate.

```


Instructions:
You are given the full RESPONSE, two claims A and B drawn from it, and a {{coupling}} relation that holds from A to B. Assuming A is TRUE, judge -- IN THE CONTEXT OF THIS RESPONSE -- whether the relation's implication about B is credible: at least plausible / more likely than not, NOT whether it is certain.

- entailment or equivalence: given A and how the response uses it, is B at least plausibly TRUE (more likely than not)?
- contradiction or exclusive: given A and how the response uses it, is B at least plausibly FALSE (more likely than not)? (For "exclusive", A and B are exhaustive alternatives, so A being true makes B false.)
- co_necessity: A and B are a pair of which at least one holds. Given the response rules out "neither", is it at least plausible that B holds when A does NOT?

Answer with a SINGLE WORD, the very first word of your reply: Yes or No.
- Answer "Yes" if the implication is credible/plausible (even if not certain).
- Answer "No" only if the implication is implausible or the response does not actually support it.
Do not output anything before the word Yes or No.

Example 1 (response supports a near-certain entailment):
RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it meets the certified reference specification.
A: The new alloy is chemically identical to the certified reference alloy.
B: The new alloy meets the certified reference specification.
coupling: entailment
Answer: Yes

Example 2 (weak but plausible, and the response draws the link -- still Yes):
RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
coupling: entailment
Answer: Yes

Example 3 (clear contradiction stated by the response):
RESPONSE: The official statement said no one was harmed, but three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
coupling: contradiction
Answer: Yes

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
coupling: {{coupling}}
Answer: 
```

A.4 PROMPT B, BASELINE — VERBALIZED CONDITIONAL STRENGTH

2,112 characters. Retained only for the comparison reported in §8; verbalized confidence is weakly calibrated (Xiong et al., 2024).

```


Instructions:
You are given the full RESPONSE and two claims A and B drawn from it. Assume claim A is TRUE and, IN THE CONTEXT OF THIS RESPONSE, estimate how strongly A determines B under a {{coupling}} relation from A to B.

- For an entailment or equivalence coupling: how likely is B to be TRUE given A?
```

```
- For a contradiction or exclusive coupling: how likely is B to be FALSE given A? (exclusive = A and B are exhaustive alternatives, so A true forces B false.)
- For a co_necessity coupling (at least one of A, B holds): how likely is B to be TRUE when A is FALSE?

Answer with a single probability in [0, 1] to two decimals, after one short sentence of justification. A near-certain link is close to 1.00; a merely plausible link is around 0.60-0.70. End your answer with the probability wrapped in brackets on its own, exactly in the form: [p=0.NN]

Example 1:
RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it meets the certified reference specification.
A: The new alloy is chemically identical to the certified reference alloy.
B: The new alloy meets the certified reference specification.
coupling: entailment
Justification: chemical identity to a certified reference almost guarantees the specification is met, so B follows very strongly from A.
[p=0.95]

Example 2:
RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
coupling: entailment
Justification: a flawed product can plausibly drive a stock decline, but many other factors affect price, so the link is only moderate.
[p=0.65]

Example 3:
RESPONSE: The official statement said no one was harmed, but three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
coupling: contradiction
Justification: if no one was harmed, it is almost certain that B (three deaths) is false.
[p=0.93]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
coupling: {{coupling}}
```

A.5 THE TWO SENSE MENUS

The full menu offers all twelve senses of Table 3 plus NONE:

```
Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast, Concession, Alternative, Disjunction, Precedence, Succession, None
```

The restricted menu offers only the senses the evaluation corpus labels, which is the admissibility filter of §4.5 applied at elicitation time rather than after it:

```
Cause-Effect, Effect-Cause, Evidence, Restatement, Contrast, Concession, Alternative, Disjunction, Precedence, None
```

Note that the restricted menu omits Succession as well as Condition and Instantiation, leaving nine relation-bearing senses. Succession is dropped because it is the mirror of Precedence and both compile to NONE, so offering both invites an arbitrary choice on a relation that produces no factor either way. The second-generation Prompt A's body is written against this restricted menu; when it is run with the full menu, three senses are offered with no gloss and no coupling mapping in the prompt body, and we report which menu each arm used.

B THE REIFIED COHERENCE NODE, WORKED

To make Eq. 7 concrete, take the three-claim subnet consisting of the unresolved casualty conflict of Example 4 together with one entailment feeding it: claims a_7, a_{10} joined by a conflict at $p = 0.93$, and $a_4 \rightarrow a_7$ an entailment at $p = 0.65$, all priors $\frac{1}{2}$, with $\rho = \frac{1}{2}$. Writing out the eight-cell vote factor for each relation from Eq. 7 and enumerating the 2^4 worlds of (a_4, a_7, a_{10}, R) gives $\mathbb{P}(R{=}1) = 0.453$: below its prior, because both relations have violating worlds that carry appreciable mass, and the conflict's violating world $(1, 1)$ carries the most.

Two properties of the construction are worth noting. The $R = 0$ branch is flat, so $R = 0$ asserts nothing and the node cannot be driven up merely by adding relations that happen to be satisfied. And the factor is a *noisy* AND rather than a hard one: a relation with $p_r < 1$ leaves some mass on its violating worlds even when $R = 1$, so a single strongly-violated relation reduces $\mathbb{P}(R{=}1)$ without pinning it to zero. The reason we do not adopt the readout despite this reasonable behaviour is Table 5: it dips at the concession variant for the reason of Prop. 3, and it requires the free parameter ρ, violating R4.

C CORRESPONDENCE WITH MARKOV LOGIC

The coherence MRF has a natural first-order encoding (Richardson & Domingos, 2006), with $\mathbb{P}(\mathbf{a}) = \frac{1}{Z}\exp\bigl(\sum_k w_k n_k(\mathbf{a})\bigr)$ for $n_k(\mathbf{a})$ the number of true groundings of formula F_k. Evidence predicates $\mathit{Entail}(i,j)$, $\mathit{Contradict}(i,j)$, $\mathit{Equiv}(i,j)$ and $\mathit{Resolves}(h,i,j)$ carry the mined graph; $\mathit{Holds}(i)$ is the query predicate. Working out the correspondence is instructive in both directions.

C.1 THE NAIVE ENCODING IS A DIFFERENT MODEL

The obvious encoding gives each relation one weighted clause, e.g. $\mathit{Entail}(i,j) \wedge a_i \Rightarrow a_j$ with $w = \mathrm{logit}(p)$. This is *not* equivalent to Definition 1. Its induced pairwise factor is $[e^w, e^w, 1, e^w]$, which is not row-stochastic in the sense of Prop. 1: it rewards the vacuous $a_s = 0$ rows with e^w on *both* target states, so a clause is best satisfied by making its antecedent false. Brute force over the 2^{16} worlds of the recall report of §9.2 gives mean marginal support 0.455 under the naive MLN against 0.587 under the corresponding MRF, with per-claim marginals differing by up to 0.31. These are two different models, not two encodings of one.

The failure is sharpest at the mode. The all-false world satisfies every implication vacuously, so under the naive encoding the most probable configuration of any purely-implicational graph is the one in which the response asserts nothing — the maximum-a-posteriori reading of a response is that none of it holds. This is Prop. 2 appearing in a second guise, and it is the same lesson: an encoding that scores rule satisfaction rather than belief can be maximized by having nothing to satisfy.

C.2 A THREE-CLAUSE EXPANSION IS EXACTLY EQUIVALENT

Writing the log-potential in the general pairwise form $\ln\psi(a_s, a_t) = \alpha + \beta a_s + \gamma a_t + \delta\, a_s a_t$ and matching it against the with-priors tables of Table 2 gives, for π_s the source prior:

τ	α (unit)	β (source)	γ (target)	δ (conjunction)
ENTAILMENT	$\ln\frac{1-p}{\pi_s}$	0	0	$+\mathrm{logit}(p)$
CONTRADICTION	$\ln\frac{p}{\pi_s}$	0	0	$-\mathrm{logit}(p)$
EQUIVALENCE	$\ln\frac{1-p}{p}$	$\ln\frac{1-p}{p}$	0	$+2\,\mathrm{logit}(p)$

The conjunction weight is exactly $\pm\mathrm{logit}(p)$ — the interaction the implication was meant to carry all along — and the unit clause is the correction that restores row-stochasticity. With this expansion the ground Markov logic network reproduces the MRF exactly: mean support 0.5866 and $\log Z = -9.751$ for both, with zero per-claim difference to machine precision.

We establish the equivalence for the three original couplings only. EXCLUSIVE and CO_NECESSITY are not tabulated here, so the claim "the MLN encoding reproduces the MRF" should be read as scoped to the three-coupling fragment. That is a real limitation for the recall report of §9.2, whose conflicts are EXCLUSIVE; the five-claim pair of §9.1 uses CONTRADICTION throughout and so falls inside the tabulated fragment.

C.3 Which view to keep, and why both

The reason to prefer the MRF for the measure itself is Remark 1: its pairwise factor family is closed at three parameters, so the inferential vocabulary cannot quietly grow as senses are added, and every relation type is guaranteed to be a 2×2 potential. The reason to keep the Markov logic view is that the rules our model cannot express are naturally written there — a transitivity rule $\mathit{Entail}(i, j) \wedge \mathit{Entail}(j, k) \Rightarrow \mathit{Entail}(i, k)$, a concession-cancellation rule $\mathit{Resolves}(h, i, j) \wedge a_h \Rightarrow \neg \mathit{Penalize}(i, j)$ replacing the fixed discount of Eq. 3, and a double-conflict rule — and their weights could be *learned* from labelled ladders rather than set by hand. That is the route past the two limitations §10 names, and it requires an MLN solver (MC-SAT for marginals, MaxWalkSAT for the mode) rather than the bucket-elimination stack used here.

D Serialization and the variable-ordering trap

One implementation detail is worth recording because it is silent when wrong. Networks are serialized in UAI format, whose variables are identified by integer index; the index order is the sort order of $(\text{cardinality}, \text{name})$, and with uniformly binary claim variables that reduces to *lexicographic order by name*. So a_{10} sorts before a_2. Inference returns marginals by index, and mapping them back by any other convention — source order, numeric order — silently permutes marginals across claims. The score is unchanged, since Eq. 5 is a mean, but every per-claim diagnostic is then attributed to the wrong claim, and the "claim dragged below its prior" readout points at an innocent claim. The auxiliary variables of §5.2 are named with a prefix that sorts after all claim names for the same reason, so that adding them cannot shift the claim indices.